

REMEMBER WHAT YOU THOUGHT, NOT WHAT YOU SAID: SELF-DISTILLED MEMORY ADMISSION FOR LLM AGENTS

Anonymous authors

Paper under double-blind review

ABSTRACT

An agent must choose what to store before a future question arrives, and discarding a useful inference can cost it the answer. We find that the ordering of an importance report—a model’s yes/no judgment of what to remember—follows its candidate pool’s provenance structure: across six models, useful-concept AUC is 0.322–0.427 on the standard pool, where the useful concept is usually unstated, and 0.658–0.736 when the useful concept is equally likely to be stated or unstated. Self-Distilled Admission (SDA) ranks candidates with a frozen model’s workspace readout and output likelihood, labels the top two, and trains one adapter on those labels through the model’s own importance prompt. On the primary statedness-balanced pool, SDA improves Qwen3-8B’s answer recall by 10.7 percentage points over its own report and 5.2 points over the 1–10 importance rating at two stored concepts, and exceeds each model’s own report in 54 of 54 model–seed–budget comparisons. All three seeds reach exact McNemar $p < 0.05$ in 17 of 18 model–budget cells. The served Qwen3-8B adapter achieves 50.1% recall on the standard pool at 242 candidates/s. These results make admission a trainable component of agent memory, using a model’s own computation to preserve useful inferences for future questions.

1 INTRODUCTION

An LLM agent must decide what to store before it knows the next question. The most useful memory is often an inference the passage never states: a passage about a landmark can identify a city without naming it, and remembering that city preserves the link to a later question about the local language. Once the passage leaves the context, the answer depends on which inferences were kept.

Memory systems admit candidates through importance judgments the model expresses, such as ratings and admit-or-discard decisions (Park et al., 2023), or through criteria built from what the context supports (Zhang et al., 2026; Zhang & Li, 2026). All of these read what the passage says. An inference the passage leaves unstated has the least support from any of them.

We find that the ordering of a model’s importance report follows provenance: whether the passage states a candidate. Across six models, useful-concept AUC is 0.322–0.427 on the standard pool, where the useful concept is usually the unstated candidate, and 0.658–0.736 on the statedness-balanced pool, where it is stated in half the episodes. Naming the useful concept in an otherwise unchanged episode raises mean workspace-readout AUC from 0.563 to 0.956 and the report’s from 0.373 to 0.504 (Fig. 1). Admission criteria built on context support show the same ordering: single-pass ConsistencyGate scores correlate +0.90 with statedness on the statedness-balanced pool, and its own threshold admits 26.7% of useful candidates on the standard pool.

The model’s internal computation carries the inference its report does not express. Self-Distilled Admission (SDA) distills that computation into the admission decision: a frozen model ranks candidates by its workspace readout and output likelihood, labels its two highest-ranked candidates, and trains one low-rank adapter (Hu et al., 2021) on its own importance prompt. Supervision comes from the model’s processing of the passage and candidates, with no future question, reward or external teacher. The adapter runs in a standard serving engine and scores only admission requests.

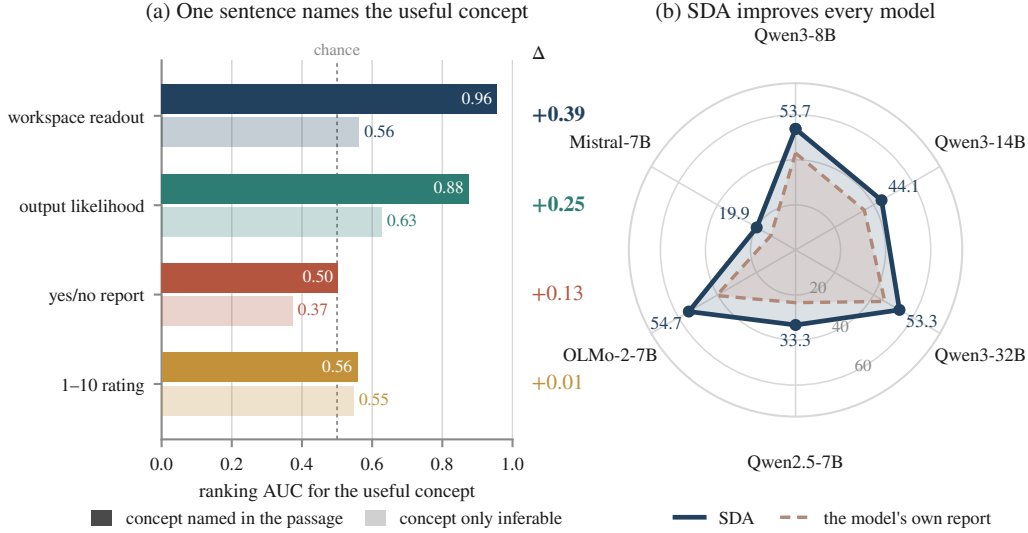


Figure 1: **Naming a concept moves model computation far more than importance reports express; Self-Distilled Admission (SDA) turns that computation into admission decisions.** (a) Matched episodes, candidates, labels and questions differ by one sentence naming the otherwise inferable concept. On the standard Fresh-2 pool, horizontal bars show six-model mean pooled ranking AUC (0–1; dashed chance line, 0.5): named concepts are solid, inferable concepts pale. Workspace readout rises from 0.563 to 0.956 (+0.39), output likelihood from 0.628 to 0.876 (+0.25), the yes/no report from 0.373 to 0.504 (+0.13) and the 1–10 rating from 0.548 to 0.558 (+0.01); changes appear at right. (b) Radar axes show percent recall at two stored concepts on the primary, statedness-balanced pool for Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B, OLMo-2-7B and Mistral-7B. The solid SDA polygon (three-seed mean) exceeds the dashed polygon of each model’s own importance report on every axis, averaging a 9.8-point gain.

On the statedness-balanced pool, SDA exceeds each model’s own importance report in all 54 model–seed–budget comparisons across six models and exceeds the 1–10 importance rating in 52 of 54 (Table 1); for Qwen3-8B at two stored concepts, SDA improves recall by 10.7 points over the model’s report and 5.2 points over the rating. Admission criteria from ConsistencyGate, A-MAC and Nemori fall below the untrained report on this pool, and SDA exceeds these criteria by 14.1 to 27.8 points at two stored concepts (Table 2). On a battery where only the passage identifies the useful concept, removing the workspace readout from SDA’s teacher lowers recall by 1.9 points at one stored concept across five models (Table 4). The served Qwen3-8B adapter reaches 50.1% recall on the standard pool at 242 candidates per second (Table 6).

Our contributions are:

- **Provenance governs importance judgments.** The model’s own importance report and admission criteria built on context support order memory candidates by provenance; the model’s internal computation tracks the inference the passage leaves unstated. Controlled pools that balance statedness and naming isolate this effect.
- **Self-distilled admission.** SDA is a plug-and-play admission adapter distilled from the model’s own internal rankings and trained without future questions, rewards or an external teacher. It serves in a standard engine and applies the learned decision to memory admission requests.
- **Gains across models and memory settings.** Evaluation spans six models from three families, three seeds, three memory budgets and five candidate pools, with gains over the model’s own importance report, the 1–10 importance rating and admission criteria from published methods. Ablations show that both internal signals in SDA’s teacher contribute to recall.

2 BACKGROUND AND RELATED WORK

Importance and memory admission. Generative Agents combines an importance rating with recency and relevance for retrieval (Park et al., 2023). MemoryBank, MemGPT, Mem0 and A-MEM provide persistence, memory operations and organization (Zhong et al., 2024; Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025). Admission methods use externally supplied utility signals: A-MAC combines LLM assessment with textual support and memory attributes (Zhang et al., 2026), a multi-factor value model optimizes evidence retention (Chen & Cheng, 2026), and Causal Memory Intervention scores answers to a known request under memory interventions (Srivastava, 2026). SDA derives admission targets from the host model’s own computation before a future request is available. Our provenance controls show how importance reports prioritize passage content during this decision. The controlled admission tests complement long-conversation evaluations such as LoCoMo and LongMemEval (Maharana et al., 2024; Wu et al., 2025).

Learned memory management. Learned memory management spans admission, control of memory operations and distillation of experience. ConsistencyGate (Zhang & Li, 2026) is a write-time admission gate that filters candidates by self-consistent support in the source context. Nemori (Ma et al., 2026) distills interaction sequences into memory using prediction error and shares the motivation that heuristic importance scores encode designer intuition. MemCon (Jiang et al., 2026) models retrieval, consolidation and forgetting as a learned control policy. AgeMem (Yu et al., 2026) exposes memory operations as tool actions trained with reinforcement learning. TMEM (Ren et al., 2026) writes distilled experience into low-rank weights instead of an external store. Agent Memory Distillation (Kim et al., 2026) transfers a larger teacher agent’s trajectories into structured memory for a smaller student. The survey of Hu et al. (2025) provides broader context for these memory systems. SDA addresses the admission decision for a single passage before any future request, with supervision from the same model’s internal rankings rather than support in the context, prediction error, task reward or a larger teacher. Its self-distillation occurs within one model, distinguishing it from teacher-to-student memory distillation.

Internal signals and representation readouts. EM-LLM, SirLLM and Titans use surprisal, entropy and gradient-based surprise for memory (Fountas et al., 2025; Yao et al., 2024; Behrouz et al., 2025). SDA turns internal candidate rankings into admission supervision. The workspace readout builds on the logit lens and J-lens (nostalgebraist, 2020; Gurnee et al., 2026); tuned lenses, Future Lens and Patchscopes also decode internal representations (Belrose et al., 2023; Pal et al., 2023; Ghandeharioun et al., 2024). SDA uses these measurements offline to train a served language interface.

Self-distillation and internal information. Knowledge distillation transfers teacher predictions to a student, including between models of the same architecture (Hinton et al., 2015; Furlanello et al., 2018). SDA transfers a frozen model’s rankings into its own importance prompt.

Internal states can predict correctness beyond expressed confidence (Kadavath et al., 2022; Azaria & Mitchell, 2023; Orgad et al., 2025), and stated reasoning can omit influences on an answer (Turpin et al., 2023). Salience can likewise be poorly reported (Trienes et al., 2025); introspection training teaches models to predict their own behavior (Binder et al., 2025). SDA makes this transfer serve a prospective decision: admit a concept before the question that uses it.

3 CONSTRUCTING THE ADMISSION SETTING

The setting makes a useful inference compete with concepts drawn from the passage for a fixed number of memory slots. We build each episode around a bridge concept: a piece of information that links the passage to a future answer. The episode supplies a passage, an ordered candidate set, utility labels, and a future question with its answer. Only the passage and candidates reach the admission policy.

3.1 FROM A PASSAGE TO A USEFUL MEMORY

Decoupled requires two inference steps: recover the bridge from the passage, then combine it with general knowledge to answer the future question. GPT-4.1 authors a passage about a named person together with the bridge, question and answer. The answer is excluded from the passage and candidate set; the question connects to the passage through the person’s name. The author supplies vivid surface distractors and neutral passage concepts as negative utility labels, with the intended bridge as the positive label. Mechanical filters check candidate strings, bridge and answer exclusion, and question overlap. Deduplication removes repeated concepts, and a blind alias audit checks that recovering the bridge still leaves an inference to the answer (App. A).

Decoupled episode

Passage. Jibril had always wanted to gaze up at the world’s highest outdoor observation deck. Early one morning, he took a swift elevator to level 148 of the Burj Khalifa, joining a crowd of eager tourists. Below him, the city sparkled with the first rays of the desert sun. After snapping several panoramic photos, he browsed the sleek gift shop for souvenirs.

Candidates, in supplied order. elevator, desert, gift, observation, dubai, tourists.

Useful concept. dubai. **Surface distractors.** desert, tourists. **Neutral concepts.** elevator, gift, observation.

Future question. What language was Jibril most likely to hear from local staff during his visit?

Answer. arabic.

Figure 2: A complete Fresh-2 episode. The passage identifies a landmark from which the city can be inferred. Retaining *dubai* preserves the city-to-language link needed by the future question. Utility labels, question and answer are visible here for explanation; admission sees the passage and candidate strings.

With two memory slots, the city in Figure 2 competes with the elevator, gift shop and scenery. **Explicit** instead states the useful content, while **Compositional** requires a multi-hop answer and permits passage anchors in the question. **Fresh-1** and **Fresh-2** repeat the Decoupled construction with new passages and bridges absent from the training batteries. Each battery is frozen before measurement.

The author uses the bridge and question to identify the intended memory; the policy predicts its value from the passage and candidates alone. Changing admission changes the remembered concepts while keeping the future question and reader fixed.

3.2 CONTROLS NAMED FOR WHAT THEY CHANGE

Mention and position. The **stated-bridge control** appends *The passage also mentions bridge*. The **position control** puts that sentence before the passage. The **distractor-named control** substitutes a distractor for the bridge in the appended sentence. Candidates, utility labels, question and answer stay fixed, isolating the named concept and sentence position.

Competition for memory. Our primary **statedness-balanced pool** starts from the 363 held-out Fresh-2 episodes in their frozen order. Counting from the first episode, odd positions keep the passage unchanged, providing the inferred-concept condition; even positions append the naming intervention’s sentence *The passage also mentions bridge*. Here *bridge* is replaced by the useful concept, providing the stated-concept condition. Every episode retains its original candidates, including the stated distractors, and gains four unstated distractors at odd positions or three at even positions. These are the first four or three additional candidates, in order, from the pool with unstated distractors described below. Both alternations follow position in the frozen order, with no sampling. The resulting 363 episodes average 7.93 candidates; 13.3% of stated candidates and 12.0% of unstated candidates are useful. The probe question and answer, original utility labels, frozen reader and blind judge remain the same; added candidates receive negative utility labels. A statedness-only rule ranks near chance: AUC is 0.515 pooled and 0.516 within episodes.

The **naming-balanced pool** names four candidates in every episode, using five sentence templates at varying positions. Named candidates are useful 12.5% of the time, against a 12.6% base rate. The pools control different cues: statedness in the statedness-balanced pool and intervention naming in the naming-balanced pool. Their residual regularities point in opposite directions: intervention naming favors the useful concept in the statedness-balanced pool, whereas overall statedness favors distractors in the naming-balanced pool.

Two further pools test competition for the same memory budget, starting from the standard Fresh-2 pool’s 4.43 candidates per episode. The **pool with unstated distractors** adds four confusable concepts from the bridge’s semantic category, proposed by an author model and filtered for absence from the passage and original pool. The **enlarged pool** adds further concepts from other episodes, reaching 12.41 candidates on average. Both additions preserve the original passage, candidates, utility labels and future question while increasing the alternatives that admission must rank.

Context, authorship and bridge category. **Shuffled context** replaces the passage with another episode’s passage while retaining its candidates and labels. The **alternate-author benchmark** uses GPT-4o to write episodes with an implied useful concept. The **entity-bridge benchmark** uses GPT-5.6 to place a named-entity bridge among scene-object distractors. Together these variants test dependence on the paired passage, authorship and the kind of concept admitted (App. G.3).

4 EVALUATION SETUP

Given a passage x and ordered candidates $C = (c_1, \dots, c_m)$, admission stores $K \subseteq C$ with $|K| \leq k$. The future question q and answer a are hidden. When q arrives, the frozen reader sees only q and K ; the original passage has left its window. Every policy receives the same candidates and budget. We measure three successive outcomes: candidate ordering, retained information and answer correctness.

Ranking within the episode. Within-episode area under the receiver operating characteristic curve (AUC) is the fraction of useful–distractor pairs ordered correctly, with half credit for ties, averaged across episodes. For useful concept b_i in episode i , score s and n episodes,

$$A_{\text{episode}} = \frac{1}{n} \sum_{i=1}^n \frac{1}{|C_i| - 1} \sum_{c \in C_i \setminus \{b_i\}} [\mathbf{1}\{s(b_i) > s(c)\} + \frac{1}{2}\mathbf{1}\{s(b_i) = s(c)\}]. \quad (1)$$

It measures the ordering among concepts that actually compete for memory. Pooled AUC instead compares candidates across the entire battery and summarizes score discrimination. Retention at budget k connects this ordering to the admitted set. For the provenance analysis, we replace utility labels with statedness labels and report *provenance AUC*: discrimination of stated from unstated candidates under a case-insensitive stem match against the passage.

Retention of the useful concept. Bridge retention is the fraction of episodes whose stored set contains the construction-labelled useful concept:

$$B_k = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{b_i \in K_i\}. \quad (2)$$

It directly measures what survives the budget; correct answering additionally depends on the reader and the other stored concepts. An oracle-label policy prioritizes the intended bridge and provides a reference for this construction.

Downstream recall. Recall is the fraction of future questions answered correctly from stored concepts by the frozen reader. It measures the downstream effect of admission for a fixed reader. A policy can rank the bridge above most distractors yet drop it at a small budget; a retained bridge can still lead to an incorrect answer. Reporting all three outcomes locates these distinct transitions from score to memory to answer. The reader answers greedily, with Qwen3 thinking disabled. A blind semantic judge receives only the question, gold answer and response; it accepts a committed correct answer or exact equivalent. A separate string-match grader checks agreement (App. D.1). Question-only and whole-passage readers establish how much the answer can be recovered before admission and with the complete original input.

Comparisons. Baselines are the untrained yes/no importance report, a 1–10 importance rating, reversed yes/no report, admit what the passage omits, omission then report, wording variants, the provenance-adjusted report, the combined teacher used directly, random selection, output likelihood, the workspace readout, and students trained from either teacher signal alone. All receive the passage with the future question hidden (App. B.4). Paired AUC differences use 4,000-draw whole-episode bootstraps and 95% intervals; recall comparisons use exact McNemar tests on paired answers and seed-averaged differences with paired 95% bootstrap intervals (Table 16). Trained results give three-seed means and sample standard deviations. Bold table entries identify SDA.

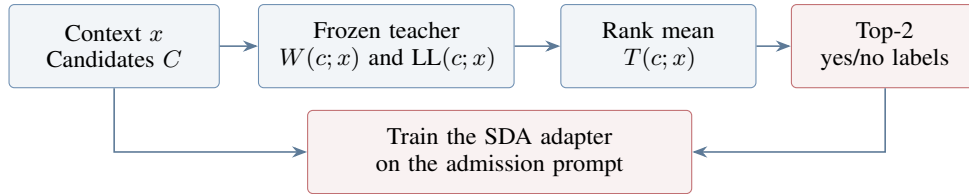
Evaluation scale. The principal diagnostic comparison covers eight models from four families: Qwen2.5-7B, OLMo-2-7B, Mistral-7B and Qwen3 at 1.7B, 4B, 8B, 14B and 32B (Qwen Team, 2025; Walsh et al., 2025; Jiang et al., 2023; Yang et al., 2025). The extended grid contains 23 checkpoints across those families and GPT-2, including base and post-training stages (App. G.2). SDA trains three seeds for each of six models: Qwen3-8B, 14B and 32B and the other three 7B families, totaling 18 principal training runs. Each supplies its own teacher, adapter and frozen reader.

Training and validation contain 175 and 45 episodes, drawn from 88 Explicit episodes and 132 implied-answer episodes, including the 57-episode alternate-author benchmark. In an implied-answer episode, the passage cues an unstated useful concept that itself supplies the answer; Decoupled instead requires a further inference from that concept. Development uses Decoupled (68 episodes), Compositional (52), and two separate 68-episode pools that add confusable or unrelated unstated distractors to Decoupled. Fresh-1 and Fresh-2 contain 316 and 363 episodes; the entity-bridge benchmark contains 42. Fresh-2 passage and pool controls reuse its 363 episodes. Budgets are $k = 1, 2, 3$; the main comparison uses the statedness-balanced pool, with $k = 2$ reported across all six models. The paired Fresh-2 grading audit totals 13,068 answers across the 18 SDA-report comparisons; both graders evaluate every answer.

5 SELF-DISTILLED ADMISSION

SDA distills a frozen model’s candidate rankings into yes/no admission judgments, training the importance interface to retain useful inferences (Fig. 3).

(a) Training: offline, once



(b) Deployment: standard serving

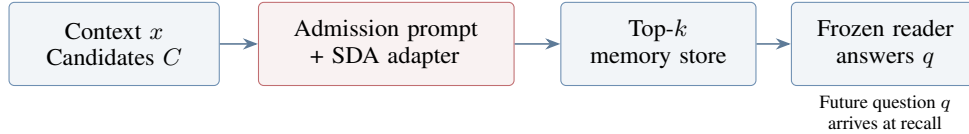


Figure 3: **SDA training and deployment.** (a) Both the context x and the candidate set C enter the frozen teacher. Workspace and likelihood ranks produce top-2 labels for training the admission adapter. (b) The host supplies context and candidates to the trained importance prompt, stores the top- k concepts, and later queries its frozen reader. The future question enters only at recall.

5.1 A TEACHER FROM THE MODEL’S OWN COMPUTATION

Workspace readout. Let $h_\ell(x) \in \mathbb{R}^d$ be the residual state at the final context position after transformer layer ℓ , with L layers in total. Final normalization LN and unembedding W_U give vocabulary logits $z_\ell(x) = W_U \text{LN}(h_\ell(x))$. Writing $\text{rank}_\ell(t) = 1 + |\{v : z_{\ell,v} > z_{\ell,t}\}|$, the workspace readout is the peak reciprocal rank across layers and first-token forms:

$$W(c; x) = \max_{1 \leq \ell \leq L} \max_{t \in T(c)} \frac{1}{\text{rank}_\ell(t)}. \quad (3)$$

Here $T(c)$ contains the distinct first tokens of the candidate’s bare and leading-space forms, each in original and capitalized form. The readout uses one forward pass over x and final-position states from all layers. It measures how readily the model’s vocabulary decoder recovers a concept during context processing (nostalgebraist, 2020; Gurnee et al., 2026).

Teacher used in the main evaluation. The teacher averages workspace and output-likelihood ranks. The latter scores the complete candidate: $\text{LL}(c; x) = \log p_{\mathcal{M}}(c \mid x)/|c|$, where $|c|$ is the candidate’s token length. Workspace decodability and output likelihood have different scales. We map each to mid-rank percentiles within the episode, assigning tied values their mean rank (App. B), and average them:

$$T(c; x) = \frac{1}{2} (P_W(c; x) + P_{\text{LL}}(c; x)). \quad (4)$$

Here P_S denotes the percentile under readout S . The two highest-ranked candidates receive the binary label $\tilde{y} = 1$, corresponding to *yes*; every other candidate receives $\tilde{y} = 0$, corresponding to *no*. Candidate order breaks score ties. The frozen model generates every teacher label from the passage and candidates.

5.2 DISTILLING THE ADMISSION JUDGMENT

The importance prompt $u(c, x)$ asks whether c is one of the most important concepts to remember for possible future questions, requesting a single yes or no (App. B.5). Qwen3 uses thinking-disabled prompts. Let Y, N be the distinct first tokens of the yes/no case and leading-space forms. The admission score is

$$s_\theta(c; x) = \log \sum_{t \in Y} p_\theta(t \mid u(c, x)) - \log \sum_{t \in N} p_\theta(t \mid u(c, x)), \quad V_\theta(c; x) = \sigma(s_\theta(c; x)). \quad (5)$$

The untrained model supplies the report baseline V ; SDA ranks candidates by the adapter-enabled score s_θ .

Only a LoRA adapter is optimized; base parameters remain frozen. The training objective is cross-entropy on the teacher’s binary labels, implemented as next-token prediction of the target word:

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{(x,c) \in \mathcal{D}} \log p_\theta(\tau(\tilde{y}(c; x)) \mid u(c, x)). \quad (6)$$

Here τ maps a positive label to the word *yes* and a negative label to the word *no*. The target probability is normalized over the vocabulary, and prompt tokens are masked from the loss (App. B.2).

Training and selection. We train for two epochs using a rank-16 adapter and select the earliest checkpoint maximizing in-distribution validation admission AUC. Optimization uses teacher labels; validation uses construction-assigned utility labels. Mean recall across development batteries and training seeds, scored by the development run’s automatic grader, selects the combined teacher used by SDA. The configuration is fixed before Fresh-2 (App. C).

5.3 ADMISSION THROUGH STANDARD SERVING

At deployment, SDA makes one yes/no forward pass per candidate and stores the top- k by s_θ . The adapter is attached only to admission requests in a standard serving engine such as vLLM (Kwon et al., 2023); prefix caching reuses the shared passage. Serving reads the admission prompt’s output probabilities. Other requests, including the frozen reader’s answer generation, use the unmodified model. The host stores the admitted concepts and manages retrieval. Candidate order breaks score ties.

Table 1: Recall on the statedness-balanced pool: 363 episodes, 7.93 candidates per episode, with alternating inferred- and stated-concept conditions. Recall in percent at $k = 2$, blind semantic grading; all three training seeds are shown. The 1–10 importance rating (Park et al., 2023) is the strongest untrained policy on this pool for Qwen3-8B. Bold identifies SDA. p bounds the exact McNemar test against each model’s own yes/no report.

Model	Yes/no report	1–10 rating	Workspace readout	SDA (3 seeds)	p vs. report	Oracle labels
Qwen3-8B	43.0	48.5	43.0	53.7 / 54.8 / 52.6	$\leq 10^{-5}$	54.3
Qwen3-14B	35.3	40.0	36.6	44.9 / 44.1 / 43.2	$\leq 10^{-4}$	46.6
Qwen3-32B	45.7	43.5	46.0	52.1 / 54.3 / 53.4	$\leq 10^{-3}$	54.8
Qwen2.5-7B	23.4	26.4	24.2	35.3 / 30.6 / 34.2	$\leq 10^{-3}$	31.7
OLMo-2-7B	40.5	47.7	45.2	54.8 / 54.5 / 54.8	$\leq 10^{-7}$	54.8
Mistral-7B	12.7	16.5	16.2	19.3 / 20.4 / 20.1	$\leq 10^{-3}$	16.8

6 RESULTS

6.1 IMPORTANCE REPORTS FOLLOW THE POOL’S PROVENANCE STRUCTURE

The report ranks useful concepts above chance on the statedness-balanced pool, and the workspace readout ranks them higher still. Across six models, their pooled AUCs are 0.658–0.736 and 0.719–0.812, respectively. A rule that reads only whether the passage states a candidate ranks near chance: useful-concept AUC is 0.515 pooled and 0.516 within episodes, over all 363 episodes. The within-episode value is the relevant test: candidates compete for memory inside each episode. On the standard pool, report AUC is 0.322–0.427 and provenance AUC is 0.582–0.688. The report’s ordering follows the provenance structure of the pool it is given. Naming the bridge raises mean workspace AUC from 0.563 to 0.956 and report AUC from 0.373 to 0.504 (Table 21); naming a distractor lowers both on Qwen3-8B (App. F.4). Among unstated candidates alone, the report ranks usefulness above the workspace readout across all six models (App. F.1); the full-pool ordering also reflects competition across provenances.

Omission directly identifies the useful concept, the sole unstated candidate, in 95% of standard-pool episodes. At $k = 2$, admit what the passage omits and omission then report both reach 53.4% recall; reversed yes/no report reaches 46.6%. Balancing provenance changes these recalls to 32.0%, 39.1% and 29.8%: the cue that favored the bridge now also favors distractors. All three rank below the report in recall at $k = 2, 3$; omission then report exceeds it at $k = 1$ (Tables 2, 15 and 16). On the standard pool, SDA trails omission then report by 9.5 points at one stored concept ($[-13.7, -5.3]$); the difference at two is -3.4 points ($[-7.0, +0.4]$). On the statedness-balanced pool, SDA exceeds this rule by 14.6 points at two stored concepts ($[+9.4, +19.7]$) (Tables 16 and 10). The provenance-cued wording that says the passage may never mention the useful concept follows this pattern: Qwen3-8B AUC is 0.537 when the concept is inferred, 0.417 when named, and 0.621 on Explicit (App. F.3).

6.2 SDA RETAINS USEFUL CONCEPTS AND IMPROVES LATER RECALL

SDA exceeds each model’s own report in all 54 model–seed–budget comparisons on the statedness-balanced pool. At $k = 2$, Qwen3-8B rises from 43.0% to 53.7/54.8/52.6%; all six models improve in every seed (Table 1). Across budgets, all three seeds reach exact McNemar $p < 0.05$ in 17 of 18 model–budget cells. The 1–10 importance rating is the strongest untrained admission policy on this pool for Qwen3-8B, and it is evaluated for all six models at all three budgets. SDA exceeds it in 52 of 54 model–seed–budget comparisons; the two exceptions are Qwen3-14B at three stored concepts, where two of three seeds reach 44.9% against the rating’s 45.1%.

On the statedness-balanced pool, Qwen3-8B SDA reaches 40.5/53.7/51.8% recall at $k = 1, 2, 3$ (Table 2). The untrained admission policies reach 35.8/48.5/50.7% for the 1–10 rating, 31.4/44.6/47.7% for the implied-content wording, 36.6/39.1/39.9% for the provenance-adjusted report, and 36.9/44.9/46.3% for the combined teacher used directly. Direct output likelihood reaches 34.7/41.9/41.6%. The contributions of the teacher’s two signals are ablated in §6.4.

Table 2: Admission policies on the statedness-balanced pool, Qwen3-8B, 363 episodes. Recall is in percent under blind semantic grading; trained policies average three seeds. A statedness-only rule ranks near chance (AUC 0.515 pooled, 0.516 within episodes). ConsistencyGate uses its paper’s sampled form (verbatim Appendix C prompt, $K = 5$, mean parsed decimals) and single-pass form (softmax over yes/no token log-probabilities). A-MAC uses its temperature-zero utility factor; novelty and recency are constant within an episode, and a five-fold cross-validated combination of utility, confidence and type-prior factors gives no gain over utility alone. Nemori’s prediction-error criterion is implemented as candidate surprisal under the passage. Oracle labels prioritize the intended bridge. Table 10 gives paired differences for every row.

Admission policy	$k = 1$	$k = 2$	$k = 3$
<i>Importance judgments of the untrained model</i>			
Yes/no importance report	30.6	43.0	45.5
1–10 importance rating (Park et al., 2023)	35.8	48.5	50.7
Implied-content wording	31.4	44.6	47.7
<i>Admission criteria from published methods</i>			
ConsistencyGate, sampled (Zhang & Li, 2026)	24.0	35.0	40.8
ConsistencyGate, single pass (Zhang & Li, 2026)	24.5	35.0	42.4
A-MAC utility (Zhang et al., 2026)	30.0	39.7	44.9
Prediction error (Ma et al., 2026)	16.5	25.9	30.3
<i>Provenance rules</i>			
Provenance-adjusted report	36.6	39.1	39.9
Reversed yes/no report	23.7	29.8	33.1
Admit what the passage omits	28.6	32.0	36.4
Omission then report	36.9	39.1	39.9
<i>Internal signals used directly</i>			
Output likelihood used directly	34.7	41.9	41.6
Combined teacher used directly	36.9	44.9	46.3
SDA (mean of 3 seeds)	40.5	53.7	51.8
Oracle labels	47.4	54.3	51.5

At $k = 2$, paired SDA gains are +5.2 points over the rating ([+1.7, +8.8]), +9.1 over implied-content wording ([+5.7, +12.7]), +14.6 over the provenance-adjusted report ([+9.5, +19.8]), and +8.8 over the direct combined teacher ([+3.9, +13.9]). These are seed-averaged differences with 95% intervals from 4,000 paired episode-bootstrap draws; Table 10 gives all budgets.

We implemented ConsistencyGate (Zhang & Li, 2026), A-MAC utility (Zhang et al., 2026) and Nemori’s prediction-error criterion (Ma et al., 2026) from their papers, admitting top- k candidates under the same budget protocol (Table 2). On the primary, statedness-balanced pool at two stored concepts, recall is 35.0% for both ConsistencyGate forms, 39.7% for A-MAC utility and 25.9% for prediction error: all three methods fall below the model’s own yes/no importance report (43.0%), while SDA reaches 53.7%. Paired, seed-averaged SDA gains are +18.7 points ([+14.1, +23.6]) over sampled ConsistencyGate, +18.7 ([+14.0, +23.5]) over single-pass ConsistencyGate, +14.1 ([+9.3, +19.0]) over A-MAC utility and +27.8 ([+22.5, +33.2]) over prediction error, with 95% bootstrap intervals over episodes from 4,000 draws. Every one of the 12 intervals for these methods in Table 10 excludes zero; the smallest gain is +6.9 ([+2.6, +11.1]), over A-MAC utility at three stored concepts. ConsistencyGate and A-MAC utility track whether the passage states a candidate: correlations on the statedness-balanced pool are +0.90 for single-pass ConsistencyGate, +0.85 for sampled ConsistencyGate and +0.75 for A-MAC utility. On the standard pool, ConsistencyGate’s own 0.7 threshold admits 26.7% of useful candidates against 70.8% of all candidates; its within-episode useful-concept AUC is 0.133 for single-pass scoring and 0.190 for sampled scoring. These independently proposed admission methods exhibit the same provenance ordering identified in importance reports.

Table 3: Reader references and candidate proposal on 363 Fresh-2 episodes, Qwen3-8B. Recall is in percent under blind semantic grading. The reader references answer from the question alone or from the whole passage, without admission. A context-only proposer supplies 8.0 candidates per episode and includes the useful concept in 1.4% of episodes; admission from these candidates stores two concepts, and SDA shows each seed. Host-supplied standard-pool recall for each policy is in Table 15.

Condition	Recall
<i>Reader references</i>	
Question only	22.9
Whole passage	63.9
<i>Admission from proposed candidates</i>	
Yes/no importance report	33.6
Oracle labels	34.4
SDA (3 seeds)	35.5 / 33.3 / 33.1

A fitted provenance offset reproduces the development pool’s omission regularity, while SDA improves admission across candidate pools. Adding a single offset for stated candidates, fitted on development episodes, raises the report’s development AUC from 0.304 to 0.978 and lowers its statedness-balanced AUC from 0.683 to 0.575. The fitted offset orders every unstated candidate ahead of every stated candidate, reproducing omission then report. Their recalls match at $k = 2, 3$; at $k = 1$, they are 36.6% and 36.9%. The offset learns which provenance the benchmark rewards; SDA learns a passage-and-candidate admission judgment.

For Qwen3-8B at two stored concepts, retention separates stated and inferred conditions: the yes/no report retains the useful concept in 46.4% versus 33.0% of episodes, the workspace readout in 92.8% versus 35.7%, and SDA in 76.8/81.8/72.4% of stated episodes versus 64.8/64.8/64.3% of inferred episodes across three seeds. SDA improves inferred-concept retention over the workspace readout and retains stated concepts. The 173 inferred-condition episodes contain no sentence naming any candidate. On these episodes, SDA reaches 37.4/50.1/46.6% recall at $k = 1, 2, 3$ against 24.9/40.5/37.0% for the model’s own yes/no report, gains of 12.5 ([+6.7, +18.7]), 9.6 ([+3.7, +15.6]) and 9.6 ([+4.4, +15.0]) points. SDA exceeds the 1–10 rating by 5.6 points at two stored concepts ([+0.8, +10.8]) on these inferred-condition episodes. Differences average three seeds with 95% bootstrap intervals over episodes from 4,000 draws.

Useful-concept retention carries the recall gain. On the standard pool, Qwen3-8B within-episode AUC rises from 0.326 to 0.679–0.688 and retention from 29.8% to 68.6–69.7% (Table 17). Replaying SDA’s two selections on the 249 episodes where it retains the bridge, then replacing that bridge with its next-ranked candidate, drops recall from 54.6% to 37.8%: a paired 16.9-point decrease, with 59 answers lost against 17 gained (exact McNemar $p = 1.4 \times 10^{-6}$). The retained concept supplies information the frozen reader uses.

Across six models on the naming-balanced pool, SDA exceeds the model’s own yes/no report in 17 of 18 model–budget cells, and the remaining cell differs by 0.1 points (Table 7). SDA exceeds the 1–10 rating in all twelve model–budget cells at one and two stored concepts; at three stored concepts, SDA and the rating are within 2.6 points in every model. Qwen3-8B within-episode AUC reaches 0.715–0.725 across seeds versus 0.566 for the report, and retention at $k = 2$ reaches 53.2–54.0% versus 24.0% (Table 8). The admission gain persists when naming carries no utility information.

On the standard pool, all 18 runs exceed their own report, with a mean 9.6-point gain and exact McNemar $p \leq 0.026$ (Tables 14 and 16). Both semantic and string-match grading recover all 18 gains (App. D.1).

Reader references and candidate proposal. Question-only and whole-passage readers reach 22.9% and 63.9% recall on Qwen3-8B, bracketing SDA’s 50.0% on the standard pool. With a context-only proposer, SDA reaches 35.5/33.3/33.1% across three seeds at $k = 2$ (Table 3; proposal prompt in App. B.5).

Table 4: Paired passages share candidates but have different useful concepts and questions; only the passage identifies the useful concept. Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B and OLMo-2-7B provide 1,810 observations per budget. Entries are SDA minus the student of the teacher with one signal removed, averaged over three seeds and pooled over models, in points. Recall is blind semantic answer grading; retention indicates admission of the episode’s useful concept. The 95% intervals use 4,000 bootstrap draws resampling pairs, carrying each pair’s rows across all five models together.

Budget	Without the workspace readout		Without the output likelihood	
	Retention	Recall	Retention	Recall
$k = 1$	+7.3 [+6.1, +8.5]	+1.9 [+1.0, +2.8]	−1.4 [−2.9, +0.2]	+1.3 [+0.3, +2.3]
$k = 2$	+4.6 [+3.6, +5.7]	+0.6 [−0.2, +1.4]	+0.1 [−1.3, +1.4]	+1.6 [+0.7, +2.5]
$k = 3$	+3.0 [+2.0, +3.9]	+1.1 [+0.4, +1.8]	+1.7 [+0.6, +2.7]	+1.4 [+0.5, +2.3]

6.3 RECALL GAINS PERSIST WITH UNSTATED DISTRACTORS

SDA improves recall when the useful inference competes with other unstated concepts. All 54 model–seed–budget comparisons exceed their model’s report on the pool with unstated distractors (Table 12). On Qwen3-8B, direct output likelihood reaches 30.6/39.1/40.5% recall at $k = 1, 2, 3$, against SDA’s 37.6/50.7/51.6%; on the enlarged pool, the corresponding recalls are 28.1/32.0/33.9% versus 38.6/51.5/52.4% (Tables 15 and 13). At $k = 2$, SDA exceeds the report by 9.6 points (paired 95% CI [5.5, 13.8]), the rating by 4.1 points ([0.6, 7.6]), and omission then report by 5.8 points ([0.8, 10.8]; Table 16). With 12.41 candidates in the enlarged pool, its corresponding gains are 11.0 points ([6.7, 15.2]), 4.7 points ([0.9, 8.4]), and 6.6 points ([1.6, 11.6]). At one stored concept, SDA trails omission then report by 8.4 points on the pool with unstated distractors ([−12.7, −4.2]) and by 5.5 points on the enlarged pool ([−9.6, −1.5]) (Table 16). At two and three stored concepts, SDA leads every measured non-oracle baseline on both pools (App. E.1).

6.4 TEACHER COMPOSITION AND THE ADMISSION OBJECTIVE

We ablate SDA’s teacher by removing either the workspace readout or the output likelihood and training the same adapter on the resulting labels, holding episodes and step budget fixed. On the paired-passage battery where only the passage identifies the useful concept, removing either signal from SDA’s teacher lowers answer recall.

Each candidate set appears twice, with two passages and their own questions. The useful concept differs within each pair, so both episodes include their partner’s useful concept and only the passage identifies which is useful. The 181 pairs are scored on Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B and OLMo-2-7B, yielding 1,810 observations per budget (Table 4; per-model results: Table 29). At one, two and three stored concepts, the teacher without the workspace readout incurs recall costs of +1.9 ([+1.0, +2.8]), +0.6 ([−0.2, +1.4]) and +1.1 ([+0.4, +1.8]) points; retention costs are +7.3 ([+6.1, +8.5]), +4.6 ([+3.6, +5.7]) and +3.0 ([+2.0, +3.9]) points. The teacher without the output likelihood incurs recall costs of +1.3 ([+0.3, +2.3]), +1.6 ([+0.7, +2.5]) and +1.4 ([+0.5, +2.3]) points; retention costs are −1.4 ([−2.9, +0.2]), +0.1 ([−1.3, +1.4]) and +1.7 ([+0.6, +2.7]) points. Costs are SDA minus the trained ablation, averaged over three seeds and pooled over models. The 95% intervals use 4,000 bootstrap draws resampling pairs and carrying each pair’s rows across all five models together.

The replaced-passage test scores each episode with its own passage and another episode’s passage, keeping candidates, labels and question fixed. Table 5 reports retention for Qwen3-8B on 363 standard-pool episodes with one stored concept. SDA’s retention drops from 48.3% to 27.4%, a decrease of 20.9 points ([+14.3, +27.4]). For the student of the teacher without the workspace readout, retention drops from 42.8% to 39.9%, or 2.9 points ([−3.5, +9.2]). For the student of the teacher without the output likelihood, it drops from 46.6% to 20.1%, or 26.5 points ([+20.5, +32.6]). Without the workspace readout, the student’s ordering barely changes when the passage changes.

Table 5: Passage replacement on the standard Fresh-2 pool, Qwen3-8B, 363 episodes. Each episode is scored twice, with its own passage and with another episode’s passage, keeping candidates, labels and question fixed. The SDA row and the two rows named by the signal removed from SDA’s teacher are trained adapters; the two untrained rows score candidates directly. Retention is the rate at which the single admitted candidate is useful, at a budget of one stored concept. Students average three seeds. Drops are own-passage minus replaced-passage retention, with paired 95% bootstrap intervals over episodes from 4,000 draws.

Admission policy	Own passage	Replaced passage	Drop
SDA	48.3	27.4	+20.9 [+14.3, +27.4]
Without the workspace readout	42.8	39.9	+2.9 [−3.5, +9.2]
Without the output likelihood	46.6	20.1	+26.5 [+20.5, +32.6]
Workspace readout, untrained	26.2	14.3	+11.8 [+7.4, +16.2]
Yes/no report, untrained	16.0	20.4	−4.4 [−9.9, +1.1]

The primary pool combines five models and 1,815 observations per budget, with each episode moving together across models in the bootstrap. The teacher without the output likelihood incurs recall costs of +1.6 ([+0.7, +2.6]) and +1.7 ([+0.8, +2.7]) points at one and two stored concepts. The teacher without the workspace readout incurs costs of +0.7 ([−0.2, +1.6]), +0.6 ([−0.2, +1.4]) and +0.3 ([−0.4, +1.1]) points at one, two and three stored concepts. Table 27 gives per-model recall; Table 28 gives paired intervals.

On the standard Fresh-2 pool at $k = 2$, SDA reaches 50.0% recall, versus 45.7% for the combined teacher used directly and 40.2–40.5% for the validation-selected log-odds objectives (Tables 32 and 33).

Training the importance interface adds recall beyond changing its wording. The *implied-content wording* is the strongest measured Qwen3-8B prompt baseline that permits inferences while leaving the useful concept’s provenance unspecified. It raises standard-pool report AUC from 0.321 to 0.462 and recall at $k = 1, 2, 3$ from 28.1/40.8/41.6% to 29.2/41.3/44.9%. Holding this wording fixed for training and evaluation, SDA reaches 37.1/50.7/51.0%, exceeding the prompt baseline at every budget (exact McNemar $p \leq 6.9 \times 10^{-5}$; Table 19). Passage-only wording SDA reaches 37.6/50.0/52.2% (Tables 15 and 16). The matched wording comparison isolates the additional gain from learning the admission judgment.

With teacher, episodes, adapter and step budget fixed, validation within-episode admission AUC starts at 0.657 and reaches 0.824 at the final step under SDA’s vocabulary-normalized token objective. Binary cross-entropy on the yes/no log-odds ends at 0.250, 0.333 and 0.301 across three seeds at learning rate 10^{-5} ; validation selects the untrained checkpoint in every run. At a step size five times smaller (2×10^{-6}), selection returns a trained checkpoint, whose standard-pool recall at $k = 2$ is 40.2%, against 50.0% for SDA’s token-level objective (Table 33).

6.5 STANDARD SERVING PRESERVES RECALL

Standard serving preserves SDA recall while scoring 242 candidates per second on one A100-80GB (Table 6). The served Qwen3-8B adapter achieves 50.1% standard-pool recall, within 0.3 percentage points of the reference selections. Paired disagreements are two versus three episodes (exact McNemar $p = 1.0$); top-2 memories agree on 91.7% of episodes and admission AUC differs by at most 0.001. Batched Qwen3-8B serving on the standard pool with prefix caching on one A100-80GB reaches 242 candidates/s (4.1 ms per candidate), against 290 candidates/s (3.4 ms) for the report, over 1,607 candidates after warm-up (Table 6). The adapter serves admission requests; the frozen model answers from the admitted memory.

Table 6: Admission serving on Qwen3-8B, standard pool, one A100-80GB and 1,607 candidates after warm-up. Amortized time divides batched wall time by candidate count, with prefix caching. Served top-2 selections agree with the reference on 91.7% of episodes; admission AUC differs by at most 0.001.

Admission scoring	Time (ms/candidate)	Candidates/s	Hidden states
Yes/no importance report	3.4	290	not needed
SDA adapter	4.1	242	not needed

7 IMPLICATIONS

For builders of agent memory systems. Treat admission as a trainable decision in the memory pipeline. A host can attach SDA to admission requests while keeping its reader frozen. Qwen3-8B serves the standard pool at 242 candidates/s, making the learned selection available through standard serving (§6.5).

For builders of models. Train the importance interface to use rankings available in the model’s own computation. SDA improves development admission AUC at every tested Qwen3 training scale and seed while workspace AUC changes by at most 0.0046 (App. G.4), making prospective memory selection a target for self-distillation.

For practitioners evaluating memory. Compare admission policies on the same episodes, with the reader and budget fixed, useful-concept provenance balanced, and unstated distractors present. Report within-episode ranking, retention split by provenance, and frozen-reader recall to connect candidate scores to stored information and answers.

8 CONCLUSION

An importance report’s ordering follows its candidate pool’s provenance structure. SDA turns internal rankings into admission supervision and exceeds each model’s own report in all 54 comparisons on the statedness-balanced pool. Standard serving preserves recall. Memory systems can train admission before the future question arrives and retain the inferred links that answering will need.

9 LIMITATIONS

The authored episodes evaluate host-supplied candidate admission for one later question; end-to-end long-conversation memory also requires proposal, revision, retrieval and deletion. Construction labels identify an intended bridge, not the optimal memory for every reader. Stem matching approximates provenance, and semantic grading uses a model judge. Serving measurements cover batched Qwen3-8B scoring on the standard pool on one A100-80GB. Full-candidate readout measurements cover Qwen3-8B on the standard pool. The objective ablation evaluates ranking and recall rather than probability calibration, with three seeds at learning rate 10^{-5} and one at each smaller learning rate. The 54 model–seed–budget comparisons share episodes and measure consistency across settings, not 54 independent replications.

REFERENCES

- Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP*, 2023.
- Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time. In *Advances in Neural Information Processing Systems*, volume 38, pp. 125925–125962, 2025. doi: 10.52202/085713-3786.
- Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.

- Felix J. Binder, James Chua, Tomek Korbak, Henry Sleight, John Hughes, Robert Long, Ethan Perez, Miles Turpin, and Owain Evans. Looking inward: Language models can learn about themselves by introspection. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=eb5pkwIB5i>.
- Zhibao Chen and Qian Cheng. Learning what to remember: A cognitively grounded multi-factor value model for agentic memory. *arXiv preprint arXiv:2606.12945*, 2026.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. In *ECAI 2025*, volume 413 of *Frontiers in Artificial Intelligence and Applications*, pp. 2993–3000. IOS Press, 2025. doi: 10.3233/FAIA251160.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018. doi: 10.48550/arXiv.1803.05457.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. doi: 10.48550/arXiv.2110.14168.
- Zafeirios Fountas, Martin A. Benfeghoul, Adnan Oomerjee, Fenia Christopoulou, Gerasimos Lampouras, Haitham Bou-Ammar, and Jun Wang. Human-inspired episodic memory for infinite context llms. *ICLR*, 2025. arXiv:2407.09450.
- Tommaso Furlanello, Zachary C. Lipton, Michael Tschannen, Laurent Itti, and Anima Anandkumar. Born-again neural networks. In *Proceedings of the 35th International Conference on Machine Learning*, 2018.
- Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models. *Proceedings of ICML*, 2024.
- Wes Gurnee, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, and Jack Lindsey. Verbalizable representations form a global workspace in language models. Transformer Circuits Thread, 2026. URL <https://transformer-circuits.pub/2026/workspace/index.html>.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021. arXiv:2009.03300.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021. doi: 10.48550/arXiv.2106.09685.
- Yuyang Hu, Shichun Liu, Yanwei Yue, Guibin Zhang, et al. Memory in the age of AI agents. *arXiv preprint arXiv:2512.13564*, 2025.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Eric Hanchen Jiang, Zhi Zhang, Yuchen Wu, Levina Li, Dong Liu, Xiao Liang, Rui Sun, Yubei Li, Edward Sun, Haozheng Luo, Zhaolu Kang, Aylin Caliskan, Kai-Wei Chang, and Ying Nian Wu. Memory as a controlled process: Learned adaptive memory management for LLM agents. *arXiv preprint arXiv:2607.13591*, 2026.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

- Taeil Kim, Kangsan Kim, and Sung Ju Hwang. Agent memory distillation: Empowering small LLM agents with hierarchical teacher memory. *arXiv preprint arXiv:2608.07169*, 2026.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023.
- Wenquan Ma, Jiayan Nan, and WenLong Wu. What deserves memory: Adaptive memory distillation for LLM agents. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 34789–34812, 2026.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- nostalgebraist. Interpreting gpt: the logit lens. LessWrong, 2020.
- Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=KRnsX5Em3W>.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- Koyena Pal, Jiuding Sun, Andrew Yuan, Byron C. Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, pp. 548–560, 2023. doi: 10.18653/v1/2023.conll-1.37.
- Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. *Proceedings of UIST*, 2023.
- Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.
- Tao Ren, Weiyao Luo, Hui Yang, Rongzhi Zhu, Xiang Huang, Yuchuan Wu, Bingxue Chou, Jieping Ye, Jiafeng Liang, Yongbin Li, and Yijie Peng. Scaling self-evolving agents via parametric memory. *arXiv preprint arXiv:2606.04536*, 2026.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. doi: 10.48550/arXiv.2402.03300.
- Saksham Sahai Srivastava. Causal intervention-based memory selection for long-horizon llm agents. *arXiv preprint arXiv:2605.17641*, 2026.
- Jan Trienes, Jörg Schlötterer, Junyi Jessy Li, and Christin Seifert. Behavioral analysis of information salience in large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 23428–23454, 2025. doi: 10.18653/v1/2025.findings-acl.1204.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. *Proceedings of NeurIPS*, 2023.
- Evan Pete Walsh, Luca Soldaini, et al. 2 OLMo 2 furious (COLM’s version). In *Second Conference on Language Modeling*, 2025. URL <https://openreview.net/forum?id=2ezugTT9kU>. Extended technical report: arXiv:2501.00656.
- Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. Longmemeval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations (ICLR)*, 2025.

- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-MEM: Agentic memory for LLM agents. In *Advances in Neural Information Processing Systems*, volume 38, pp. 17577–17604, 2025. doi: 10.52202/085713-0593.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. doi: 10.48550/arXiv.2505.09388.
- Yao Yao, Zuchao Li, and Hai Zhao. Sirllm: Streaming infinite retentive llm. *Proceedings of ACL*, 2024. arXiv:2405.12528.
- Yi Yu, Liuyi Yao, Yuexiang Xie, Qingquan Tan, Jiaqi Feng, Yaliang Li, and Libing Wu. Agentic memory: Learning unified long-term and short-term memory management for large language model agents. *arXiv preprint arXiv:2601.01885*, 2026.
- Guilin Zhang, Wei Jiang, Xiejiashan Wang, Aisha Behr, Kai Zhao, Jeffrey Friedman, Xu Chu, and Amine Anoun. Adaptive memory admission control for LLM agents. In *ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems*, 2026. URL <https://openreview.net/forum?id=mmdqUrEY24>.
- Yan Zhang and Shibo Li. ConsistencyGate: Preventing memory contamination in LLM agents via self-consistency admission control. *arXiv preprint arXiv:2607.22962*, 2026.
- Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. MemoryBank: Enhancing large language models with long-term memory. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(17):19724–19731, 2024. doi: 10.1609/aaai.v38i17.29946.

A BENCHMARK CONSTRUCTION AND LEAKAGE AUDIT

A.1 PASSAGES, CONCEPTS AND FUTURE QUESTIONS

The primary author model is GPT-4.1. Each episode contains a passage, candidate concepts with construction-assigned utility labels, and a future question and answer. Explicit places useful content in the passage. **Implied-answer** training episodes cue an unstated useful concept that itself supplies the answer, including episodes from the alternate-author benchmark. The extended tables use this name for that training construction. Decoupled requires an unstated bridge to support a further inference: the answer is absent from the passage and candidate set, and the question shares no passage content word except a protagonist’s name. Compositional requires a multi-hop inference and permits question anchors from the passage. Decoupled and Compositional are development batteries used for configuration selection and excluded from gradient training.

Mechanical validation checks candidate strings, bridge absence, answer exclusion and the Decoupled question’s lexical overlap. Concept deduplication removes repeated concepts across retained episodes. The leakage audit additionally checks full lexemes, inflected answers and question anchors. A blind GPT-6-Astra alias audit receives the future question, gold answer and bridge. It rejects identity, morphological or derivational variants, synonyms, translations, abbreviations and containment that trivially yields the answer. For example, *caterer/catering* is an alias, whereas *Madrid/Spanish* requires a further inference. The audit records the verdict and exclusion reason for each episode.

A.2 EPISODE-AUTHORING PROMPT

GPT-4.1 receives the system instruction, “You are a careful dataset designer for a cognitive-science experiment on LLM memory.” The following prose transcription preserves the generation constraints; the output-format instruction requests one structured record per episode.

Episode-authoring prompt

Design a batch of diverse test episodes about silent bridge reasoning for a memory experiment. Each episode has a passage of four to eight sentences about a named person; invent a distinctive first name, different in every episode, and vary the domain widely. A question asked later, out of context, refers to the episode only through the person’s name and requires an unstated bridge fact inferable from the passage, followed by common general knowledge. Supply a short correct answer of one to three words and five to seven candidate concepts, each a common lowercase word with a utility category and a short reason.

Exactly one candidate is the useful bridge concept, and its word must be absent from the passage. The question must share no passage content word except the person’s name: someone who never read the passage must be unable to answer it. For a passage about Elena wandering the Plaza Mayor, ask what language Elena should have brushed up on for her trip, without naming the plaza in the question. The answer must differ from every candidate and be absent from the passage; combining the bridge with general knowledge yields the answer. For example, the Plaza Mayor cues *madrid*, which supports *spanish*. A reader remembering only the bridge plus the question must be able to answer.

Include two or three vivid, emotional or surprising distractors that occur literally in the passage but are useless for the question, and one or two neutral concepts that occur in the passage. Every candidate must be a single common English lowercase word. Do not reuse concepts or person names across episodes.

Generation uses temperature 1.0 and batches of up to five episodes. Mechanical filters and the alias audit apply after generation.

A.3 FROZEN BATTERIES AND REPLICATION BENCHMARKS

Fresh-1 and Fresh-2 use the Decoupled construction prompt and mechanical filters, followed by deduplication and disjointness filtering. A context match after lowercasing and trimming against a prior battery causes exclusion. A bridge matching a construction-labeled training concept also causes exclusion. Fresh-2 additionally excludes contexts appearing in Fresh-1. Shared answer categories, such as languages and currencies, are permitted. The retained batteries contain 316 and 363 episodes, respectively. The alternate-author benchmark repeats the inferred-concept construction with a different author model. The entity-bridge benchmark places a named-entity bridge among scene-object distractors.

A.4 RECONSTRUCTING THE REPORTED CONTROLS

All pool interventions traverse the 363 held-out Fresh-2 episodes in their frozen order and preserve the question, answer and original utility labels. Added candidates receive negative utility labels. Within an episode, original candidates precede additions, whose recorded order is retained. The frozen control archive records the passages, ordered additions, named candidates and context permutation; replay uses these recorded draws. The standard pool contains 1,607 candidates, 4.43 per episode. Statedness uses the case-insensitive stem match specified in §4.

A worked source episode. The first episode reads: “Yara compared renewal quotes from different insurers online and reviewed her accident history. She called the hotline to ask clarifying questions about premium increases. After considering the coverage limits and deductibles, Yara finally entered her payment details to confirm the policy.” Its candidates, in order, are *insurer*, *deductible*, *history*, *policy*, *insurance*; *insurance* is useful. The question is “Which professional would be most relevant for Yara to contact regarding her recent purchase?” and the answer is *agent*. The second episode is Jibril’s visit in Figure 2.

Stated-bridge, position and distractor-named controls. The exact sentence form is “The passage also mentions *concept*.” The stated-bridge control appends this sentence with the useful concept; the position control prepends it. The distractor-named control appends it with the first non-useful candidate in the original order. Thus Yara’s stated-bridge control adds “The passage also mentions insurance.” after the passage, the position control puts that sentence before “Yara compared,” and the distractor-named control instead appends “The passage also mentions insurer.” These three controls keep all five candidates and their order unchanged.

Pool with unstated distractors. An author model proposes four single-word concepts from the useful concept’s semantic category, asking for alternatives a careless reader might confuse with it but that the passage does not imply. The instruction requires lowercase words absent from the passage and a comma-separated response. We retain the response order, discard a proposal that matches an original candidate or occurs as a case-insensitive substring of the passage, and append the accepted concepts. The frozen pool contains 1,450 additions: four in 361 episodes and three in episodes 24 and 263 after filtering, giving 3,057 candidates in total (8.42 per episode). For Yara, the ordered additions are *warranty*, *mortgage*, *pension*, *annuity*; her passage and five original candidates stay fixed. The development confusable-distractor control applies the same procedure to Decoupled. The unrelated-distractor control instead takes bridges from subsequent episodes, as described for the enlarged pool below.

Statedness-balanced pool. Odd positions, counting from one, keep the original passage; even positions append “The passage also mentions *concept*.” with the useful concept substituted. Every episode keeps its original candidates and takes the first four accepted additions at odd positions or the first three at even positions from the pool with unstated distractors, preserving their order. This yields 2,877 candidates, 7.93 per episode. Yara occupies position one, so her passage is unchanged and her pool is the five original candidates followed by *warranty*, *mortgage*, *pension*, *annuity*. Jibril occupies position two: append “The passage also mentions dubai.” to Figure 2’s passage and append *doha*, *riyadh*, *muscat* to its six candidates. The original question and answer remain fixed in each case.

Naming-balanced pool. Use the same ordered candidate lists as the statedness-balanced pool, but start from the original passages. Name four candidates per episode. At odd positions exclude the useful concept and sample four other candidates uniformly without replacement; at even positions include the useful concept and sample three others. Use one Python standard pseudorandom generator seeded with zero, process episodes in frozen order, prepend the useful concept to the sampled list when included, and shuffle each selected list before constructing sentences. Sampling and shuffling consume the same continuing random stream. The five sentence forms, in rotation, are “The passage also mentions *concept*.”, “*concept* is relevant here.”, “Note that *concept* comes up as well.”, “There is also *concept* to consider.”, and “*concept* features in this as well.”

Number episodes and insertions from zero for the placement rule. The first selected candidate uses the sentence form at the episode’s position in the five-form cycle; each subsequent candidate advances one form. Strip trailing full stops from the original passage, split at full-stop–space boundaries, trim each segment and restore one full stop to each. Before each insertion, count the current sentence list, including earlier additions. The insertion boundary is the remainder of the episode index plus three times the insertion index upon division by that count plus one. Boundary zero precedes the passage; the final boundary follows it. Insertions thus advance through an evolving sentence list rather than all occurring at its end.

For Yara, the shuffled selection is *pension, annuity, insurer, policy*. The additions are “The passage also mentions pension.”, “annuity is relevant here.”, “Note that insurer comes up as well.”, and “There is also policy to consider.” Their successive insertion boundaries are zero, three, zero and two. The final passage begins with the insurer, pension and policy sentences, then Yara’s first two original sentences, then the annuity sentence, then her final original sentence. There are 1,452 naming interventions, 181 of which name the useful concept: a named candidate is useful 12.5% of the time, against the 12.6% base rate among 2,877 candidates. Questions, answers and candidate order stay fixed.

Enlarged pool. Combine the standard candidate list, the accepted same-topic additions, and four bridges from other episodes. To choose the latter, start with the next episode in frozen order, advance cyclically, and accept a bridge only if it is absent from the passage, absent from the original candidate list and distinct from earlier accepted bridges. Stop after four accepted bridges. Concatenate the three lists in this order and retain the first occurrence of each concept. The resulting 4,506 candidates average 12.41 per episode. Yara’s pool is *insurer, deductible, history, policy, insurance, warranty, mortgage, pension, annuity, dubai, church, kanji, doctor*. Her passage, question and answer remain unchanged. The unrelated-distractor control uses just the standard candidates and the four cross-episode additions; for Yara these are *dubai, church, kanji, doctor*.

Shuffled context and bridge replacement. Shuffled context uses the archived episode permutation, replacing each passage while preserving its own candidates, utility labels, question and answer. The first episode receives the passage at position 105, about Lars skating on slick tracks with thin-bladed boots; its candidates remain Yara’s five concepts and its answer remains *agent*. The bridge-replacement control acts on SDA’s recorded top-two selections only when they contain the useful concept: retain the other selection and replace the useful concept with the next-ranked candidate, then replay the same frozen reader and question. For a selection of *dubai, elevator* with *gift* next, the replayed memory is *gift, elevator*; Jibril’s question and *arabic* answer stay fixed. This example illustrates the replacement rule; the measured 249 replays use the recorded selections (§E.2).

Candidate proposal and reader references. The context-only proposer receives the original passage and the instruction in Appendix B.5; the future question and bridge are hidden. Its parsed output replaces the supplied candidate list in proposal order. For Yara the resulting candidates are *online, accident, hotline, premium, coverage, deductibles, payment, policy*, so *insurance* is unavailable to admission. At two stored concepts, the same frozen reader answers from each policy’s selected pair. The question-only reference gives that reader Yara’s question with no memory; the whole-passage reference gives the same question and her original three-sentence passage. Neither reference performs admission. The wording controls change only the admission instruction, whose five exact forms appear in Appendix F.3. Authorship and entity-bridge controls repeat the episode construction with the authors and concept categories specified in Appendix G.3.

SDA improves ranking and retention under the naming-balanced construction (Table 8). Across Qwen3-8B seeds, within-episode AUC is 0.715–0.725, versus 0.566 for the report and 0.636 for the workspace readout. Useful-concept retention at $k = 2$ is 53.2–54.0%, versus 24.0% and 40.8%, respectively.

Table 7: Naming-balanced pool recall across 363 episodes. Every episode names four candidates using five sentence templates; a named candidate is useful at the base rate. Recall is reported in percent under blind semantic grading at one, two and three stored concepts for six models. Report is the model’s own yes/no importance report; Rating is the 1–10 importance rating (Park et al., 2023); SDA is the mean of three seeds.

Model	$k = 1$			$k = 2$			$k = 3$		
	Report	Rating	SDA	Report	Rating	SDA	Report	Rating	SDA
Qwen3-8B	27.6	35.3	36.4	42.7	45.2	46.2	46.3	47.1	47.0
Qwen3-14B	21.8	30.3	33.7	33.3	36.4	40.8	41.3	43.2	41.2
Qwen3-32B	31.7	30.0	41.1	43.2	43.0	48.9	50.7	51.8	53.4
Qwen2.5-7B	9.4	11.6	13.3	21.2	25.6	28.3	28.6	33.3	33.0
OLMo-2-7B	24.8	30.6	37.0	39.4	43.0	46.7	44.6	52.1	49.5
Mistral-7B	4.7	5.8	9.9	9.6	13.2	16.0	19.8	21.5	23.7

Table 8: Candidate ranking and useful-concept retention on the naming-balanced pool, Qwen3-8B, 363 episodes and 2,877 candidates. Every episode names four candidates using five sentence templates at varying positions. A named candidate is useful 12.5% of the time, against a 12.6% base rate. AUC uses construction utility labels; retention is in percent.

Admission policy	Pooled	Episode	B_1	B_2	B_3
	AUC	AUC			
Yes/no importance report	0.570	0.566	9.4	24.0	42.7
Workspace readout	0.635	0.636	22.0	40.8	54.8
SDA, seed 0	0.726	0.725	36.1	54.0	67.5
SDA, seed 1	0.714	0.715	32.2	54.0	65.6
SDA, seed 2	0.728	0.725	35.5	53.2	68.0

B SDA TRAINING AND SCORING DETAILS

B.1 TEACHER CONSTRUCTION AND DATA SPLIT

Within each episode, let $r_S(c)$ be a candidate’s ascending mid-rank under readout S , with ties assigned the average of their occupied ranks. For $m > 1$ candidates, the percentile is

$$P_S(c; x) = \frac{r_S(c) - 1}{m - 1}, \quad S \in \{W, LL\}. \quad (7)$$

For a singleton pool the percentile is one half. Averaging the percentiles yields the teacher in Eq. (4); descending teacher score and then candidate order determine the top-2 labels. Both frozen readouts score the same episode and candidate before these labels are constructed.

A source group is one author-and-construction battery: Explicit (88 episodes), implied-answer from the primary author (75), or implied-answer from the alternate author (57). The split is stratified by these groups and assigns whole episodes, including all their candidates, to training or validation. Within each group, episodes are sorted by a cryptographic hash of the fixed split seed and the source-qualified episode identity; the lowest 20%, rounded up, enter validation. The split seed is zero and stays fixed across optimization seeds. This gives 70/18, 60/15 and 45/12 training/validation episodes in the three groups, respectively: 175 training and 45 validation episodes in total. The teacher generates every optimization target from these passages and candidates. The selection protocol is specified in §5.2.

B.2 COMPLETE TRAINING CONFIGURATION

LoRA adapts the attention query, key, value and output projections and the feed-forward gate, up and down projections. Base parameters remain frozen. Each model supplies its own teacher rankings and receives its own admission adapter. The two highest-ranked candidates in each training episode receive positive admission targets; all remaining candidates receive negative targets. Only the target word contributes to the vocabulary-normalized cross-entropy in Eq. (6). Prompt tokens are masked. Qwen3 uses its chat template with thinking disabled.

Setting	Configuration
Adapter	Low-rank adaptation, rank 16, scaling 32; attention and feed-forward projections
Optimizer	AdamW; constant learning rate 1×10^{-5}
Batching	Batch size 1; gradient accumulation over 4 batches
Training duration	Two epochs; 414 optimizer steps
Sequence length	At most 1024 tokens
Data	175 training and 45 validation episodes; source-stratified, disjoint episodes
Targets	Two highest teacher ranks positive; all others negative

B.3 MEASUREMENT CONVENTIONS

The yes/no report in every main table is scored in log space over the yes and no token sets. The wording comparison uses its own common scorer for all five wordings, so its rows are comparable to one another under that aggregation: Mistral-7B’s passage-only wording gives AUC 0.411 in the main tables and 0.440 in the wording comparison because the two scorers aggregate differently; scoring the same wording one prompt at a time also gives 0.440, identical to the batched result. On Qwen3-8B’s standard pool, first-token workspace AUC is 0.543 at the end of the passage and 0.530 when the candidate is appended and the state is read immediately before its first token; the latter is the matched baseline for full-candidate AUC 0.574. Frozen-report recall is 40.8% in the main tables and 40.5% in the objective ablation, which evaluates the untrained checkpoint through the adapter-loading path instead of the frozen path.

B.4 ADMISSION BASELINES

SDA is compared with the following baselines. The **yes/no importance report** uses the untrained model’s score V in Eq. (5), with Qwen3 thinking disabled. The **1–10 importance rating** is the expected rating under normalized probabilities of the complete answer strings. The **reversed yes/no report** admits the lowest-scoring candidates. **Admit what the passage omits** prioritizes candidates absent from the passage under the statedness test and breaks ties at random. **Omission then report** prioritizes unstated candidates and uses the yes/no importance report to order candidates within each provenance. The **provenance-adjusted report** adds a single coefficient times the candidate’s statedness indicator to the untrained report, with the coefficient fitted on development episodes and held fixed for evaluation. The **combined teacher used directly** ranks by Eq. (4) without training an adapter. The **implied-content wording** permits implied content; the **provenance-cued wording** explicitly says the passage may never mention the useful concept (App. F.3). **Random selection** samples candidates uniformly; **output likelihood** and the **workspace readout** rank candidates directly by their respective scores. Each baseline sees the passage and candidates with the future question hidden. **Oracle labels** prioritize the intended bridge; question-only and whole-passage readers provide recall references. Bold entries identify SDA throughout the comparison tables.

B.5 EXACT ADMISSION, READER, RATING AND PROPOSAL INSTRUCTIONS

The boxes below reproduce the natural-language instructions. Italic phrases stand for the supplied passage, candidate, memory list or question. Each is a user message rendered with the model’s chat template and assistant generation prefix; Qwen3 thinking is disabled.

Admission prompt, shared by SDA and the report baseline*Passage*

Based only on the passage above, is the concept “*candidate concept*” one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.

The affirmative token set contains the distinct first tokens of *yes* in lowercase, initial-capital and uppercase forms, with and without a leading space. The negative set is constructed analogously from *no*. Equation (5) gives the common report score; SDA applies the same scoring rule with its adapter enabled.

Reader prompt

Earlier you read a passage. The only things you remember from it are: *stored concepts separated by commas*.

Using only what you remember, answer concisely.

Question: *future question*

Answer:

The reader uses the frozen model with its admission adapter disabled. The question-only reference asks “Answer this question concisely.” followed by the question and answer prefix. The whole-passage reference supplies the passage, then “Answer concisely.” and the same question and answer prefix. Greedy decoding is used throughout.

Rating baseline prompt*Passage*

On a scale of 1 to 10, where 1 is purely mundane and 10 is extremely important, rate how important it is to remember the concept “*candidate concept*” in order to answer possible future questions about this passage. Answer with a single number.

Let a_r be the complete answer string for rating r , and let $u_R(c, x)$ be this prompt. Each answer receives its full sequence probability, multiplying the conditional probabilities of its tokens. The score is

$$R(c; x) = \sum_{r=1}^{10} r \pi_r(c; x), \quad \pi_r(c; x) = \frac{p_{\mathcal{M}}(a_r \mid u_R(c, x))}{\sum_{j=1}^{10} p_{\mathcal{M}}(a_j \mid u_R(c, x))}. \quad (8)$$

A rating represented by multiple tokens is scored as a complete answer.

Candidate-proposal prompt*Passage*

A memory system can store only a few single-word concepts about the passage above. List up to 8 concepts that could be worth remembering for possible future questions, including things the passage clearly implies but does not state. Answer with lowercase single words separated by commas, and nothing else.

The context-only proposer uses Qwen3-8B with greedy decoding. Distinct parsed concepts form the candidate pool. The bridge, future question and answer are withheld from proposal and carried forward for evaluation. Bridge coverage accepts case differences and a trailing plural ending.

B.6 ABLATION PROTOCOL

The teacher and optimization ablations in §6.4 use the same training episodes, adapter shape and matched step budget. The recall-reward condition samples admitted sets from the admission scores and asks the frozen reader to answer from the selected memory. Answer correctness supplies the reward. Within each rollout group, reward differences are normalized into advantages; updates use a clipped group-relative policy objective and a penalty for departure from the frozen policy (Shao et al., 2024). The future question enters the reward computation. The refinement condition applies these updates after SDA training.

C DOCUMENTED PROSPECTIVE FREEZE

The dated experiment record documents prospective freezes of Fresh-1 and Fresh-2 on 15 September 2026. It records each battery’s episode count and cryptographic digest before evaluated models read it: 316 episodes for Fresh-1 and 363 for Fresh-2. The accompanying reproducibility record retains the dated entries and verified battery digests. Fresh-1 supplied the initial frozen-battery evaluation. Fresh-2 was frozen before any evaluated model read it and after configuration selection. Its primary endpoint compares SDA with the yes/no report; a primary training seed was designated and the remaining seeds reported alongside it. Its secondary endpoint compares SDA with the rating baseline. Fresh confirmation uses the blind semantic judge in App. D.1.

D RECALL GRADING AND SERVING PROTOCOL

D.1 BLIND SEMANTIC JUDGE

The judge, GPT-6-Astra, receives the question, gold answer and reader response with no admission-policy identity. The substantive grading instruction is quoted verbatim; its output-format instruction requests a label for every supplied answer.

Grading instruction

Grade short answers to memory questions. For each item decide whether ANSWER gives GOLD as its answer.

Labels:

“correct”: the answer commits to GOLD or an exact equivalent (synonym, inflection, fuller name) as its single answer. Extra explanation is fine.

“hedged”: it offers two or more alternative answers (e.g. “X or Y”), or states the answer cannot be determined, even if GOLD is mentioned.

“incorrect”: anything else, including a different answer.

Only correct responses contribute to recall; hedged and incorrect responses count as errors. Judgments are reused for identical question, gold answer and response triples. The harness additionally grades each answer by string matching after normalization. Agreement is 85.1% across 13,068 answers; every training run exceeds its own report under both graders, with maximum exact McNemar $p = 0.026$.

Table 9: General-capability scores in percent with the adapter enabled on every token, under matched prompts. This configuration evaluates global adapter activation; SDA deployment attaches the adapter only to admission requests.

Configuration	MMLU	ARC-E	ARC-C	GSM8K
Qwen3-8B	71.6	80.4	56.2	88.4
Adapter on every token	71.5	80.7	56.4	88.6

D.2 AMORTIZED SCORING TIME AND THROUGHPUT

The timing evaluation loads Qwen3-8B on one A100-80GB and constructs an admission prompt for each of the 1,607 standard-pool Fresh-2 candidates. vLLM enables prefix caching and LoRA serving, warms the engine on an initial subset, and times batched generation over the full candidate list. Each prompt generates one token greedily; SDA attaches the adapter to the admission request, while the report baseline uses the same engine without it. Model loading is outside the timed region. Milliseconds per candidate are wall time divided by candidate count; throughput is candidate count divided by wall time. The served and reference selections are passed to the same frozen reader and semantic judge. The serving engine recovers yes/no masses from returned token log probabilities.

D.3 CAPABILITY WITH THE ADAPTER ENABLED ON EVERY TOKEN

Enabling the adapter on every token changes the four measured capability scores by at most 0.3 percentage points under matched prompts. Base and adapter scores are 71.6% and 71.5% on MMLU (Hendrycks et al., 2021), 80.4% and 80.7% on ARC-Easy, 56.2% and 56.4% on ARC-Challenge (Clark et al., 2018), and 88.4% and 88.6% on the GSM8K evaluation subset (Cobbe et al., 2021). The deployed interface attaches the adapter to admission requests.

E RECALL ACROSS EVALUATION BATTERIES AND POOLS

SDA exceeds the report in all 54 comparisons across six models, three seeds and three budgets. All seeds reach exact McNemar $p < 0.05$ in 17 of 18 model–budget cells; Qwen3-32B at $k = 3$ has maximum $p = 0.073$. Across the six models, pooled report AUC is 0.658–0.736 and workspace AUC is 0.719–0.812.

Qwen3-8B SDA improves mean recall over the report on each development and frozen battery in Table 11. The gains also occur with unstated distractors and in the enlarged pool (Tables 12 and 13). Across all six models, mean SDA recall exceeds the report and rating at every standard-pool budget (Tables 14 and 15).

E.1 CANDIDATE-POOL COMPARISONS

The standard pool rewards omission. The useful concept is the only candidate the passage never states in 95% of standard-pool episodes. The omission baseline therefore reaches 46.6%, 53.4% and 49.6% recall at $k = 1, 2, 3$, respectively; omission then report reaches 47.1%, 53.4% and 52.6%, and the reversed report reaches 35.5%, 46.6% and 49.0%. SDA’s corresponding means are 37.6%, 50.0% and 52.2%. The omission cue identifies the useful concept directly in these episodes.

Unstated distractors separate useful admission from omission. The pool with unstated distractors adds four unstated distractors per episode. Under blind semantic grading, SDA exceeds each model’s own yes/no report in 54 of 54 comparisons across six models, three seeds and budgets $k = 1, 2, 3$. All three seeds reach exact McNemar $p < 0.05$ in 17 of 18 model–budget cells; the exception is Mistral-7B at $k = 3$. At $k = 2$, the largest p across models and seeds is 0.019 (Table 12). On Qwen3-8B, direct output likelihood reaches 30.6/39.1/40.5% recall at $k = 1, 2, 3$, against SDA’s 37.6/50.7/51.6% means. SDA leads the measured non-oracle baselines at $k = 2, 3$; omission then report leads at $k = 1$ (Tables 15 and 16). At $k = 2$, the omission baseline falls from 53.4% to

Table 10: Paired SDA minus baseline recall differences on the statedness-balanced pool, Qwen3-8B, for every row of Table 2, which cites each published method (1–10 importance rating (Park et al., 2023); ConsistencyGate (Zhang & Li, 2026); A-MAC (Zhang et al., 2026); prediction error (Ma et al., 2026)). Entries average three seeds and give 95% intervals from 4,000 paired bootstrap draws over 363 episodes. Differences are in percentage points.

Baseline	$k = 1$	$k = 2$	$k = 3$
<i>Importance judgments of the untrained model</i>			
Yes/no importance report	+9.9 [+6.1, +13.8]	+10.7 [+7.0, +14.6]	+6.3 [+2.9, +9.7]
1–10 importance rating	+4.7 [+1.0, +8.3]	+5.2 [+1.7, +8.8]	+1.1 [−2.4, +4.6]
Implied-content wording	+9.1 [+5.4, +12.9]	+9.1 [+5.7, +12.7]	+4.1 [+1.0, +7.3]
<i>Admission criteria from published methods</i>			
ConsistencyGate, sampled	+16.5 [+11.6, +21.5]	+18.7 [+14.1, +23.6]	+11.0 [+6.3, +15.5]
ConsistencyGate, single pass	+16.0 [+10.8, +21.1]	+18.7 [+14.0, +23.5]	+9.4 [+4.8, +14.1]
A-MAC utility	+10.5 [+5.6, +15.3]	+14.1 [+9.3, +19.0]	+6.9 [+2.6, +11.1]
Prediction error	+24.0 [+18.9, +29.1]	+27.8 [+22.5, +33.2]	+21.5 [+16.4, +26.6]
<i>Provenance rules</i>			
Provenance-adjusted report	+3.9 [−0.9, +8.5]	+14.6 [+9.5, +19.8]	+11.8 [+6.4, +17.0]
Reversed yes/no report	+16.8 [+11.5, +22.2]	+24.0 [+18.5, +29.5]	+18.7 [+13.2, +24.3]
Admit what the passage omits	+11.8 [+6.4, +17.3]	+21.8 [+16.3, +27.2]	+15.4 [+10.0, +21.1]
Omission then report	+3.6 [−1.0, +8.6]	+14.6 [+9.4, +19.7]	+11.8 [+6.5, +17.2]
<i>Internal signals used directly</i>			
Output likelihood used directly	+5.8 [+0.6, +10.7]	+11.8 [+6.7, +16.9]	+10.2 [+5.3, +15.1]
Combined teacher used directly	+3.6 [−1.2, +8.2]	+8.8 [+3.9, +13.9]	+5.5 [+0.9, +10.0]

Table 11: Recall accuracy (%) of budgeted memory ($k=2$) on Qwen3-8B. The admission policy sees only the context and the candidate; a frozen reader answers the future question from the stored concepts. Development batteries (left) and two frozen fresh Decoupled batteries (right); Fresh-2 follows configuration selection and a documented prospective freeze on 15 September 2026. SDA: mean $\pm$ SD over three seeds. Answers graded by a blind semantic judge. *every seed exceeds the yes/no report with exact McNemar $p < 0.05$. Rating: the 1–10 importance rating (Park et al., 2023).

Admission policy	Development				Fresh (held out)	
	Decoupled $n=68$	Compositional $n=52$	Confusable $n=68$	Unrelated $n=68$	Fresh-1 $n=316$	Fresh-2 $n=363$
Yes/no importance report	35.3	46.2	35.3	36.8	34.8	40.8
1–10 importance rating	47.1	53.8	48.5	48.5	43.7	45.7
SDA	49.5 $\pm$ 2.2	54.5 $\pm$ 1.1	48.5 $\pm$ 2.5	50.0 $\pm$ 2.5	44.2 $\pm$ 0.7*	50.0 $\pm$ 1.4*
Oracle labels	51.5	65.4	51.5	51.5	50.0	54.3

35.3% and the reversed report from 46.6% to 28.9%, both below the report’s 41.0%; SDA reaches 51.0%, 51.2% and 49.9%. Every SDA seed exceeds the omission baseline at every budget, with exact McNemar $p \leq 0.097$ at $k = 1$ and $p \leq 6.5 \times 10^{-7}$ at $k = 2, 3$; every seed exceeds the reversed report with $p \leq 4.3 \times 10^{-7}$ across budgets. SDA recalls are 37.7% / 36.4% / 38.6% at $k = 1$ and 51.8% / 51.5% / 51.5% at $k = 3$; the rating baseline reaches 35.8% and 48.8%.

Naming the bridge and changing its category test reversal. On the stated-bridge control at $k = 2$, SDA reaches 55.4%, 55.6% and 52.9% recall, against 44.9% for the report and 39.9% for the reversed report. On the entity-bridge benchmark, the reversed report reaches 0.281 admission AUC, while SDA reaches 0.987, 0.980 and 0.987, with within-episode AUC 1.000. The direction required by reversal is a property of candidate construction, and is opposite on this benchmark (§6.1).

Table 12: Recall on the pool with unstated distractors (Fresh-2 with four extra candidates per episode that the passage also never states, 8.4 candidates per episode). Blind semantic grading, recall in percent, three training seeds per model; p bounds the exact McNemar test against each model’s own yes/no report across those seeds.

Model	Yes/no report	SDA (3 seeds)	p vs. report	Oracle labels
Qwen3-8B	41.0	51.0 / 51.2 / 49.9	$\leq 10^{-3}$	54.3
Qwen3-14B	32.2	40.8 / 39.4 / 39.9	$\leq 10^{-3}$	46.6
Qwen3-32B	41.0	51.0 / 50.7 / 50.1	$\leq 10^{-4}$	54.8
Qwen2.5-7B	23.7	31.4 / 28.4 / 30.6	$\leq .019$	31.7
OLMo-2-7B	38.8	52.3 / 51.0 / 51.2	$\leq 10^{-5}$	54.8
Mistral-7B	12.9	19.0 / 19.6 / 17.9	$\leq .015$	16.8

Table 13: Recall on the enlarged pool: Fresh-2 with 12.41 candidates per episode, up from 4.43. Qwen3-8B, blind semantic grading, recall in percent. Output likelihood is used directly. SDA entries give three training seeds; p bounds the exact McNemar test against the report across seeds. Oracle labels prioritize the intended bridge.

Budget	Yes/no report	Output likelihood	Workspace readout	SDA (3 seeds)	p vs. report	Oracle labels
$k = 1$	29.5	28.1	24.0	38.0 / 38.0 / 39.7	$\leq 10^{-4}$	47.1
$k = 2$	40.5	32.0	34.7	50.7 / 52.6 / 51.2	$\leq 10^{-5}$	54.3
$k = 3$	41.9	33.9	38.6	52.9 / 52.1 / 52.1	$\leq 10^{-5}$	51.5

The enlarged pool preserves gains at two and three concepts. With 12.41 candidates per episode, SDA reaches 38.0% / 38.0% / 39.7% at $k = 1$, 50.7% / 52.6% / 51.2% at $k = 2$, and 52.9% / 52.1% / 52.1% at $k = 3$ (Table 13). The rating baseline reaches 36.4%, 46.8% and 48.2%, respectively; every SDA seed leads it at every budget. The paired mean differences are +2.2 points ($[-1.3, +5.6]$), +4.7 ($[+0.9, +8.4]$), and +4.1 ($[+0.9, +7.6]$), respectively (Table 16). Omission then report has the highest recall at $k = 1$; SDA leads the measured non-oracle baselines at $k = 2, 3$. Direct output likelihood reaches 28.1/32.0/33.9%, against SDA’s 38.6/51.5/52.4% means. Random selection reaches 21.5%, 27.0% and 29.8%, the report 29.5%, 40.5% and 41.9%, and the workspace readout 24.0%, 34.7% and 38.6%.

E.2 WITHIN-EPISODE RANKING AND BRIDGE RETENTION

SDA improves both within-episode ranking and retention of the intended bridge at $k = 2$ (Table 17). Paired whole-episode bootstrap intervals compare each score with the report baseline. Conditional recall distinguishes retaining the bridge from answering the question: for the primary SDA seed, recall is 54.6% with the bridge and 41.2% without it, versus 48.2% and 37.7% for the report.

The paired replacement experiment replays the primary SDA seed’s two selections. On the 249 episodes whose memory contains the bridge, we replace only that concept with SDA’s next-ranked candidate, keeping the other selection, question and frozen reader fixed. Recall changes from 54.6% to 37.8%, a 16.9-point decrease: 59 previously correct answers become incorrect and 17 become correct (exact McNemar $p = 1.4 \times 10^{-6}$). Retaining the bridge supplies information used by the reader.

Bridge labels and reader success measure different outcomes. Qwen3-8B recall is 22.9% with the question alone, 63.9% with the whole passage, 54.3% with oracle labels and 50.0% with SDA. Oracle labels prioritize the intended bridge; the reader answers from the complete stored set and its own knowledge. On OLMo-2-7B and Mistral-7B, SDA reaches 55.4% and 19.8%, versus 54.8% and 16.8% for oracle labels.

Table 14: Memory budget across models. Recall accuracy (%) on the held-out Fresh-2 battery (363 episodes) with k stored concepts; each model admits memories with its own adapter and answers with its own frozen weights; blind semantic grading; Rating is the 1–10 importance rating (Park et al., 2023); SDA is the mean over three seeds; *every seed exceeds the yes/no report of the same model at that budget (exact McNemar $p < 0.05$).

Model	$k=1$			$k=2$			$k=3$		
	Report	Rating	SDA	Report	Rating	SDA	Report	Rating	SDA
Qwen3-8B	28.1	34.7	37.6*	40.8	45.7	50.0*	41.6	47.9	52.2*
Qwen3-14B	23.4	31.1	33.5*	32.5	39.1	40.5*	37.5	43.5	43.6*
Qwen3-32B	32.5	32.0	41.4*	41.0	44.6	50.2*	46.0	50.7	53.0*
Qwen2.5-7B-Instruct	9.6	9.6	16.8*	23.1	22.6	29.8*	29.5	35.0	35.8*
OLMo-2-7B-Instruct	27.5	33.1	48.5*	38.0	44.1	55.4*	44.4	52.6	56.9*
Mistral-7B-Instruct	5.2	9.6	12.4*	12.7	15.7	19.8*	19.3	22.0	24.2*

Table 15: Admission policies on Qwen3-8B, blind semantic grading, recall in percent. The standard pool leaves the useful concept as the only candidate the passage never states in 95% of episodes; the pool with unstated distractors adds four more candidates per episode that the passage also never states. Output likelihood is used directly. Rows above the rule are untrained baselines; dashes denote unmeasured settings.

Admission policy	Standard pool			Pool with unstated distractors		
	$k=1$	$k=2$	$k=3$	$k=1$	$k=2$	$k=3$
Yes/no importance report	28.1	40.8	41.6	28.6	41.0	41.3
1–10 importance rating (Park et al., 2023)	34.7	45.7	47.9	35.8	46.6	48.8
Reversed yes/no report	35.5	46.6	49.0	23.4	28.9	30.9
Admit what the passage omits	46.6	53.4	49.6	31.7	35.3	36.9
Random selection	28.6	44.6	46.6	—	—	—
Output likelihood	—	48.8	—	30.6	39.1	40.5
Workspace readout	29.8	44.6	49.9	26.7	39.4	44.9
SDA (mean of 3 seeds)	37.6	50.0	52.2	37.6	50.7	51.6
Oracle labels	47.1	54.3	51.5	47.1	54.3	51.5

F PROVENANCE, WORDING AND POOL-COMPOSITION MEASUREMENTS

F.1 RANKING AMONG CANDIDATES WITH THE SAME PROVENANCE

The report discriminates usefulness among unstated candidates. Restricting the pool with unstated distractors to episodes with an unstated useful concept and at least one unstated negative leaves 345 episodes, with 5.01 qualifying candidates per episode on average. Within this group, report AUC is 0.880–0.897 across six models and workspace AUC is 0.592–0.697. Every paired 95% interval for workspace minus report lies below zero (Table 18). This measurement holds provenance fixed; the full-pool comparison measures the competition between stated and unstated candidates.

F.2 THE FULL-POOL AUC CHANGES WITH THE COMPARISON PAIRS

On Fresh-2, the useful concept is stated in 5% of episodes and negative candidates in 99%; in 95% of episodes, the useful concept is the sole candidate absent from the passage. Adding unstated distractors leaves every original Qwen3-8B report score unchanged. It adds 1,450 negative candidates to the original 1,244, while retaining the same 363 positives. Useful-concept AUC against the added negatives is 0.889, compared with 0.322 against the original negatives. The pooled AUC is therefore the weighted average

$$\text{AUC}_{\text{expanded}} = \frac{1244}{2694} \text{AUC}_{\text{original}} + \frac{1450}{2694} \text{AUC}_{\text{added}} = 0.627. \quad (9)$$

Table 16: Paired SDA-baseline recall differences on three candidate pools, Qwen3-8B. Entries are seed-averaged differences in points with 95% intervals from 4,000 paired bootstrap draws over episodes. The provenance-cued wording explicitly permits a useful concept the passage never mentions. All comparisons use the same frozen reader and blind semantic judge.

Baseline	$k = 1$	$k = 2$	$k = 3$
<i>Standard pool</i>			
Yes/no importance report	+9.6 [+5.8, +13.3]	+9.3 [+5.2, +13.2]	+10.6 [+6.8, +14.1]
1–10 importance rating	+2.9 [−0.6, +6.4]	+4.3 [+0.9, +7.8]	+4.2 [+1.3, +7.2]
Provenance-cued wording	+5.4 [+1.4, +9.4]	+7.1 [+2.9, +11.4]	+6.2 [+2.8, +9.7]
Reversed yes/no report	+2.1 [−3.3, +7.4]	+3.5 [−1.9, +8.9]	+3.1 [−1.3, +7.5]
Admit what the passage omits	−8.9 [−13.1, −4.6]	−3.4 [−7.7, +1.0]	+2.6 [−1.2, +6.2]
Omission then report	−9.5 [−13.7, −5.3]	−3.4 [−7.0, +0.4]	−0.5 [−3.2, +2.4]
Random selection	+9.0 [+4.1, +14.1]	+5.4 [+0.6, +10.3]	+5.6 [+1.4, +9.8]
<i>Pool with unstated distractors</i>			
Yes/no importance report	+8.9 [+5.1, +12.8]	+9.6 [+5.5, +13.8]	+10.3 [+6.3, +14.1]
1–10 importance rating	+1.7 [−1.6, +5.0]	+4.1 [+0.6, +7.6]	+2.9 [−0.4, +6.2]
Reversed yes/no report	+14.1 [+9.0, +19.5]	+21.8 [+16.2, +27.6]	+20.8 [+15.3, +26.1]
Admit what the passage omits	+5.9 [+0.6, +10.9]	+15.4 [+10.3, +20.8]	+14.7 [+9.6, +19.8]
Omission then report	−8.4 [−12.7, −4.2]	+5.8 [+0.8, +10.8]	+8.9 [+4.0, +13.8]
Output likelihood	+7.0 [+1.7, +12.0]	+11.6 [+6.5, +16.8]	+11.1 [+6.2, +16.1]
<i>Enlarged pool</i>			
Yes/no importance report	+9.1 [+5.2, +12.9]	+11.0 [+6.7, +15.2]	+10.5 [+6.6, +14.2]
1–10 importance rating	+2.2 [−1.3, +5.6]	+4.7 [+0.9, +8.4]	+4.1 [+0.9, +7.6]
Omission then report	−5.5 [−9.6, −1.5]	+6.6 [+1.6, +11.6]	+12.1 [+7.1, +17.2]
Output likelihood	+10.5 [+5.0, +15.7]	+19.6 [+14.3, +24.9]	+18.5 [+13.3, +23.6]
Random selection	+17.1 [+12.4, +22.0]	+24.5 [+19.1, +29.9]	+22.6 [+17.4, +27.7]

Table 17: Within-episode admission ranking on the standard Fresh-2 pool (Qwen3-8B). AUC and its difference from the report have paired 95% whole-episode bootstrap intervals (4,000 draws). Retention is the percentage of episodes whose two admitted concepts contain the useful concept.

Admission score	Per-episode AUC [95% CI]	Δ vs. report [95% CI]	Bridge retained ($k = 2$)
Yes/no report	0.326 [0.286, 0.366]	<i>reference</i>	29.8
1–10 rating (Park et al., 2023)	0.608 [0.567, 0.648]	+0.282 [+0.246, +0.320]	61.2
Workspace readout	0.547 [0.511, 0.584]	+0.221 [+0.172, +0.271]	51.0
Output likelihood	0.619 [0.582, 0.654]	+0.293 [+0.242, +0.344]	59.2
SDA (seed 0)	0.682 [0.641, 0.721]	+0.356 [+0.320, +0.393]	68.6
SDA (seed 1)	0.679 [0.639, 0.718]	+0.353 [+0.316, +0.391]	69.7
SDA (seed 2)	0.688 [0.648, 0.727]	+0.362 [+0.325, +0.401]	69.2

Table 18: Useful-concept ranking restricted to unstated candidates in the pool with unstated distractors. The 345 qualifying episodes have an unstated useful concept and at least one unstated negative, averaging 5.01 candidates in the restricted set. Entries are pooled AUC; $W - V$ gives the workspace-report difference with a paired 95% whole-episode bootstrap interval from 4,000 draws.

Model	Yes/no report	Workspace readout	$W - V$ [95% CI]
Qwen3-8B	0.881	0.639	−0.242 [−0.274, −0.211]
Qwen3-14B	0.880	0.654	−0.226 [−0.257, −0.196]
Qwen3-32B	0.896	0.697	−0.199 [−0.228, −0.170]
Qwen2.5-7B	0.897	0.592	−0.305 [−0.340, −0.270]
OLMo-2-7B	0.883	0.661	−0.221 [−0.252, −0.191]
Mistral-7B	0.880	0.672	−0.208 [−0.238, −0.178]

Table 19: Implied-content wording and distillation on the standard pool, Qwen3-8B, 363 episodes. Recall is in percent. SDA averages three seeds; the final row gives seed-averaged paired differences and 95% bootstrap intervals from 4,000 draws. The same wording is used to train and evaluate the adapter.

Admission policy	$k = 1$	$k = 2$	$k = 3$
Yes/no importance report	28.1	40.8	41.6
Implied-content wording	29.2	41.3	44.9
SDA with implied-content wording	37.1	50.7	51.0
SDA minus implied-content wording	+7.9 [+4.5, +11.4]	+9.4 [+5.8, +13.1]	+6.1 [+3.5, +8.9]

The added comparisons account for 53.8% of all positive–negative pairs. They raise AUC without changing the ordering of the original candidates. At a two-concept budget, the report continues to prioritize passage content: recall is 41.0%, against SDA’s 51.0%, 51.2% and 49.9% across seeds (Table 15).

F.3 WORDING CHANGES THE IMPORTANCE ORDERING

Five matched wordings keep the episode, candidates and utility labels fixed. They ask about passage-only importance, remove that restriction, allow implied content (*implied-content wording*), state that the passage may omit the useful concept (*provenance-cued wording*), or specify a two-concept budget. The measurements cover four models on the standard pool and the pool with unstated distractors (Table 20). On Qwen3-32B, the provenance-cued wording raises standard-pool AUC from 0.398 to 0.876; the corresponding Qwen3-8B values are 0.321 and 0.537. On Mistral-7B, the implied-content wording gives 0.518, compared with 0.440 for passage-only importance. The direction and magnitude of the wording effect vary across models.

The provenance-cued wording’s ranking changes when the useful concept is stated. On Qwen3-8B, it reaches AUC 0.417 when the useful concept is named and 0.621 on Explicit. Its standard-pool recall is 32.2%, 43.0% and 46.0% at $k = 1, 2, 3$ (Table 16).

The strongest Qwen3-8B prompt baseline among wordings that permit inferences without specifying the useful concept’s provenance is the implied-content wording. Its recall rises from 28.1/40.8/41.6% to 29.2/41.3/44.9%. SDA trained and evaluated with this same wording reaches 37.1/50.7/51.0% (Table 19); all seeds exceed the wording alone at every budget, with exact McNemar $p \leq 6.9 \times 10^{-5}$. The passage-only wording SDA means are 37.6/50.0/52.2% (Table 15). Training through the implied-content wording adds recall beyond the wording change alone.

For wording j with prompt $u_j(c, x)$, the scorer is

$$s_j(c; x) = \log \sum_{t \in Y_{\text{word}}} p_{\mathcal{M}}(t \mid u_j(c, x)) - \log \sum_{t \in N_{\text{word}}} p_{\mathcal{M}}(t \mid u_j(c, x)). \quad (10)$$

The affirmative set contains the distinct token identities of *yes*, *Yes*, their leading-space forms, and *YES*, retaining a form only when the complete form is a single token; the negative set uses the analogous forms of *no*. The uppercase forms have no leading space in this scorer. All five wordings use these same sets, the answer-position logits, and thinking-disabled prompts. The main report instead uses the first token of each of six case-and-space forms, as defined in App. B.5. The two aggregations account for the Mistral measurement difference in App. B.3.

The exact question for each wording follows the passage. In the quoted instructions, *candidate* is replaced by the candidate concept.

Passage only

Based only on the passage above, is the concept *candidate* one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.

Table 20: Importance wording controls on 363 Fresh-2 episodes per pool. Entries are pooled useful-concept AUC, using the untrained model’s yes/no log odds. All wordings and models are shown.

Model	Importance wording	Standard pool	Pool with unstated distractors
Qwen3-8B	Passage only	0.321	0.628
	Unrestricted	0.391	0.663
	Implied-content wording	0.462	0.692
	Provenance-cued wording	0.537	0.726
	Two-concept budget	0.235	0.591
Qwen3-32B	Passage only	0.398	0.671
	Unrestricted	0.555	0.749
	Implied-content wording	0.656	0.802
	Provenance-cued wording	0.876	0.892
	Two-concept budget	0.327	0.633
OLMo-2-7B	Passage only	0.327	0.630
	Unrestricted	0.358	0.643
	Implied-content wording	0.420	0.675
	Provenance-cued wording	0.547	0.721
	Two-concept budget	0.301	0.614
Mistral-7B	Passage only	0.440	0.680
	Unrestricted	0.462	0.684
	Implied-content wording	0.518	0.716
	Provenance-cued wording	0.394	0.639
	Two-concept budget	0.336	0.626

Unrestricted

Is the concept *candidate* one of the most important things to remember in order to answer possible future questions about the passage above? Answer with a single word: yes or no.

Implied-content wording

Considering both what the passage above states and what it implies without stating, is the concept *candidate* one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.

Provenance-cued wording

Would storing the concept *candidate* help answer a likely future question about the passage above, even if the passage never mentions it? Answer with a single word: yes or no.

Two-concept budget

A memory system can keep only two concepts from the passage above. Should the concept *candidate* be one of them, so that a later question can be answered? Answer with a single word: yes or no.

Table 21: Matched stated-bridge control on Fresh-2. Appending a sentence naming the useful concept keeps candidates, labels and question fixed. Entries are pooled AUC. *W*: workspace readout; *LL*: output likelihood; Report: yes/no importance report; Rating: 1–10 importance rating.

Model	<i>W</i>		<i>LL</i>		Report		Rating	
	Inferred	Stated	Inferred	Stated	Inferred	Stated	Inferred	Stated
Qwen3-8B	0.543	0.952	0.613	0.859	0.322	0.515	0.568	0.581
Qwen3-14B	0.571	0.954	0.636	0.864	0.355	0.607	0.565	0.591
Qwen3-32B	0.603	0.979	0.654	0.878	0.398	0.638	0.530	0.510
Qwen2.5-7B	0.520	0.879	0.600	0.890	0.427	0.456	0.481	0.481
OLMo-2-7B	0.561	0.986	0.633	0.911	0.327	0.410	0.533	0.585
Mistral-7B	0.581	0.990	0.630	0.854	0.411	0.395	0.608	0.601
Mean	0.563	0.956	0.628	0.876	0.373	0.504	0.548	0.558

Table 22: Shuffled context on Fresh-2. Each passage is replaced by another episode’s passage, with candidates and labels fixed. Entries are pooled AUC with the paired passage and with shuffled context.

Model	Workspace readout		Output likelihood		Yes/no report	
	Paired	Shuffled	Paired	Shuffled	Paired	Shuffled
Qwen3-8B	0.543	0.433	0.613	0.604	0.322	0.485
Qwen3-14B	0.571	0.473	0.636	0.625	0.355	0.404
Qwen3-32B	0.602	0.479	0.654	0.623	0.398	0.416
Qwen2.5-7B-Instruct	0.520	0.509	0.600	0.610	0.427	0.441
OLMo-2-7B-Instruct	0.561	0.477	0.633	0.617	0.327	0.412
Mistral-7B-Instruct	0.581	0.423	0.630	0.620	0.411	0.455

F.4 MATCHED NAMING AND POSITION MEASUREMENTS

Naming the useful concept raises both internal readouts more than either importance report on Qwen3-8B, with episodes, candidates, labels and questions fixed. Workspace, output-likelihood, report and rating AUCs are initially 0.543, 0.613, 0.322 and 0.568. Appending a sentence naming the useful concept raises them to 0.952, 0.859, 0.515 and 0.581; placing the same sentence before the passage yields 0.916, 0.850, 0.582 and 0.655. Naming a distractor instead yields 0.393, 0.520, 0.249 and 0.519. Across six models, the stated-bridge control raises mean workspace AUC from 0.563 to 0.956 and mean report AUC from 0.373 to 0.504 (Table 21). The six individual report changes include Mistral-7B’s change from 0.411 to 0.395.

At $k = 2$, Qwen3-8B SDA recall on the stated-bridge control is 55.4%, 55.6% and 52.9%, against 44.9% for the report and 39.9% for the reversed report. Naming the bridge changes what an omission or reversal rule selects, while the trained module continues to retain useful information.

G EXTENDED RANKING MEASUREMENTS

G.1 INTERNAL READOUTS AND SHUFFLED CONTEXT

The report ranks useful content above the workspace readout on Explicit; the ordering reverses on Decoupled across all eight models (Table 23). Workspace rankings depend on the paired passage: shuffled context lowers workspace AUC across all six models while preserving candidates and labels (Table 22). All report measurements use Eq. (5) and the admission prompt in App. B.5.

Internal signals and the trained module use the passage. With shuffled context, workspace AUC falls across all six models, including 0.543 to 0.433 on Qwen3-8B and 0.581 to 0.423 on Mistral-7B (Table 22). Qwen3-8B likelihood AUC changes from 0.613 to 0.604 and report AUC from 0.322 to 0.485. Likelihood retains a context-independent candidate prior, while workspace ranking and the report’s preference for stated candidates depend on the paired passage. Replacing each passage with another episode’s also reduces SDA admission AUC from 0.682 to 0.526, 0.555 and 0.534 across seeds (within-episode AUC 0.540, 0.549 and 0.542), demonstrating a context-specific ranking.

G.2 BREADTH ACROSS CHECKPOINTS, FAMILIES AND STAGES

Across the 23-checkpoint, four-benchmark grid in Table 23, workspace AUC exceeds report AUC on Decoupled and Compositional at every checkpoint with a defined report, including base and post-trained stages. An entry of “—” means that the tokenizer ships no chat template, so the yes/no importance report is undefined there.

Table 23: Full predictive grid: all 23 unique checkpoints $\times$ four benchmarks (Qwen3-14B was measured in both the grid and the ladder run with identical rows and is listed once). *W*: workspace readout; *V*: yes/no importance report, the same definition used throughout the paper. All entries are pooled item-level AUCs. “—”: the tokenizer ships no chat template, so *V* is undefined. Gray rows: base (pre-trained) checkpoints; OLMo-2 Instruct checkpoints are the RLVR stage.

Checkpoint	Explicit		Implied-answer		Checkpoint	Decoupled		Compositional	
	<i>W</i>	<i>V</i>	<i>W</i>	<i>V</i>		<i>W</i>	<i>V</i>	<i>W</i>	<i>V</i>
Qwen2.5-0.5B	.5235	.4465	.5367	.4341	Qwen2.5-0.5B	.5019	.4798	.4293	.3715
Qwen2.5-0.5B-Instruct	.5211	.5496	.5158	.2916	Qwen2.5-0.5B-Instruct	.5261	.2841	.4190	.1776
Qwen2.5-1.5B	.5100	.5006	.5816	.4160	Qwen2.5-1.5B	.5903	.4052	.5451	.3879
Qwen2.5-1.5B-Instruct	.5139	.6136	.5769	.5234	Qwen2.5-1.5B-Instruct	.5768	.4637	.5489	.2936
Qwen2.5-3B	.5369	.5621	.6147	.4560	Qwen2.5-3B	.6062	.3784	.5208	.2444
Qwen2.5-3B-Instruct	.5331	.6446	.5781	.4615	Qwen2.5-3B-Instruct	.5951	.2586	.4971	.1951
Qwen2.5-7B	.5419	.5973	.6208	.4652	Qwen2.5-7B	.6121	.3730	.5152	.2771
Qwen2.5-7B-Instruct	.5376	.6792	.6341	.5856	Qwen2.5-7B-Instruct	.6423	.4897	.5156	.3472
Qwen3-0.6B	.4932	.5306	.5472	.3264	Qwen3-0.6B	.5364	.2248	.4693	.1123
Qwen3-1.7B	.4818	.5917	.5409	.3895	Qwen3-1.7B	.5607	.1961	.4162	.1812
Qwen3-4B	.5090	.6496	.5744	.4798	Qwen3-4B	.5291	.3967	.5050	.2600
Qwen3-8B	.5090	.7100	.6204	.4534	Qwen3-8B	.6545	.3872	.5466	.2079
Qwen3-14B	.5083	.7400	.6390	.5219	Qwen3-14B	.6677	.4271	.5086	.2163
Qwen3-32B	.5084	.7480	.6139	.5511	Qwen3-32B	.6919	.4782	.5532	.2774
OLMo-2-1B	.5006	—	.6308	—	OLMo-2-1B	.6001	—	.5114	—
OLMo-2-1B-SFT	.5409	.5252	.5969	.5016	OLMo-2-1B-SFT	.5725	.3716	.4760	.3064
OLMo-2-1B-DPO	.5374	.5563	.5759	.4973	OLMo-2-1B-DPO	.5802	.3799	.4764	.2786
OLMo-2-1B-Instruct	.5354	.5551	.5795	.4899	OLMo-2-1B-Instruct	.5801	.3284	.4783	.2548
OLMo-2-7B	.4806	—	.6735	—	OLMo-2-7B	.6835	—	.5453	—
OLMo-2-7B-SFT	.4959	.5827	.6827	.5369	OLMo-2-7B-SFT	.6637	.3317	.5312	.2924
OLMo-2-7B-Instruct	.4938	.6231	.6688	.5398	OLMo-2-7B-Instruct	.6664	.3529	.5341	.2944
Mistral-7B-Instruct-v0.3	.5324	.6191	.6687	.5777	Mistral-7B-Instruct-v0.3	.6713	.4700	.5962	.3242
GPT-2	.5070	—	.5388	—	GPT-2	.5331	—	.5170	—

Workspace AUC exceeds report AUC at every measured Qwen3 size on Decoupled, with non-monotonic paired changes between adjacent sizes (Table 24). From 8B to 32B, the paired change is -0.0536 (95% CI $[-0.1352, +0.0279]$).

Workspace AUC exceeds report AUC on Decoupled and Compositional at every released OLMo-2 post-training stage measured at 1B and 7B; the separation persists through SFT, DPO where available, and RLVR (Table 23).

G.3 ALTERNATE-AUTHOR AND ENTITY-BRIDGE BENCHMARKS

The report’s preference for stated candidates persists with a different author model. Across six models on the alternate-author benchmark, report AUC ranges from 0.220 to 0.384, alongside provenance AUC 0.650–0.803 (Table 25). These measurements use the same report definition as the primary batteries.

Table 24: Qwen3 dense ladder on Decoupled. Upper block: the workspace–report gap $W - V$ with a paired 95% whole-episode bootstrap interval; W and V for each checkpoint are in Table 23. Lower block: the paired change in $W - V$ between sizes, with both sizes resampled on shared episodes. *95% interval excludes zero.

Checkpoint	$W - V$ [95% CI]
Qwen3-1.7B	+0.3646* [+0.2709, +0.4522]
Qwen3-4B	+0.1324* [+0.0280, +0.2346]
Qwen3-8B	+0.2673* [+0.1670, +0.3668]
Qwen3-14B	+0.2405* [+0.1392, +0.3409]
Qwen3-32B	+0.2137* [+0.1119, +0.3177]
Size change	$\Delta(W - V)$ [95% CI]
1.7B $\rightarrow$ 4B	-0.2322* [-0.3338, -0.1321]
4B $\rightarrow$ 8B	+0.1349* [+0.0521, +0.2147]
8B $\rightarrow$ 14B	-0.0268 [-0.0949, +0.0475]
14B $\rightarrow$ 32B	-0.0268 [-0.1057, +0.0536]
8B $\rightarrow$ 32B	-0.0536 [-0.1352, +0.0279]

Table 25: Replications on two further benchmarks, measured with the same thinking-disabled yes/no report and workspace readout as the main tables. W and V are pooled AUCs; $W - V$ carries a paired 95% whole-episode bootstrap interval (4,000 draws), and * marks intervals excluding zero. The alternate-author benchmark repeats the primary construction with episodes written by a different author model; the entity-bridge benchmark makes the useful concept a named entity among scene-object distractors.

Benchmark	Model	W	V	$W - V$ [95% CI]
Alternate-author	Qwen2.5-7B	0.490	0.384	+0.106 [-0.009, +0.215]
	Qwen3-8B	0.515	0.220	+0.295 [+0.197, +0.386]*
	Qwen3-14B	0.521	0.267	+0.254 [+0.166, +0.340]*
	Qwen3-32B	0.522	0.275	+0.247 [+0.134, +0.356]*
	OLMo-2-7B	0.547	0.353	+0.194 [+0.093, +0.297]*
	Mistral-7B	0.507	0.294	+0.213 [+0.128, +0.300]*
Entity-bridge	Qwen2.5-0.5B	0.569	0.270	+0.299 [+0.211, +0.396]*
	Qwen2.5-7B	0.664	0.881	-0.217 [-0.303, -0.133]*
	Qwen3-8B	0.642	0.719	-0.076 [-0.189, +0.032]

Report success accompanies a reversal of provenance preference. On the entity-bridge benchmark, Qwen2.5-7B and Qwen3-8B report AUCs reach 0.881 and 0.719, above workspace AUCs of 0.664 and 0.642 (Table 25). Both reports prefer unstated candidates: their provenance AUCs are 0.133 and 0.289. Shuffled context reduces report AUC to 0.282 and 0.340. Qwen2.5-0.5B retains the usual preference for stated candidates, with provenance AUC 0.734 and utility AUC 0.270. The report reads the paired passage in both regimes; the direction of its provenance preference changes with candidate construction.

G.4 ADMISSION AFTER DISTILLATION ACROSS SCALES

SDA raises development admission AUC at every measured Qwen3 scale and seed (Table 26). All 18 paired 95% intervals for the gain lie above zero, while the largest absolute workspace AUC change is 0.0046. The measured improvement is concentrated in the admission interface.

H TEACHER AND OPTIMIZATION MEASUREMENTS

H.1 TEACHER COMPOSITION AND CANDIDATE READOUT

Table 27 reports primary-pool recall for all three teacher compositions; Table 28 gives their complete paired intervals.

Table 26: Admission ranking on the Decoupled and Compositional development batteries across Qwen3 scales. These batteries informed configuration choice and were excluded from gradient training. V : pooled admission AUC; ΔV : paired change with a 95% whole-episode bootstrap interval; $|\Delta W|$: largest absolute workspace change across seeds.

Model	Benchmark	V before	Seed	V after	ΔV [95% CI]	$ \Delta W $
Qwen3-8B	Decoupled	0.387	0	0.756	+0.369 [+0.310, +0.431]	0.0033
			1	0.739	+0.352 [+0.293, +0.417]	
			2	0.758	+0.371 [+0.312, +0.435]	
Qwen3-8B	Compositional	0.208	0	0.618	+0.410 [+0.325, +0.490]	0.0021
			1	0.623	+0.415 [+0.329, +0.494]	
			2	0.619	+0.412 [+0.326, +0.491]	
Qwen3-14B	Decoupled	0.427	0	0.702	+0.275 [+0.216, +0.340]	0.0029
			1	0.669	+0.242 [+0.192, +0.299]	
			2	0.680	+0.253 [+0.200, +0.312]	
Qwen3-14B	Compositional	0.216	0	0.529	+0.312 [+0.234, +0.395]	0.0029
			1	0.458	+0.242 [+0.172, +0.317]	
			2	0.481	+0.265 [+0.190, +0.346]	
Qwen3-32B	Decoupled	0.478	0	0.773	+0.295 [+0.237, +0.356]	0.0037
			1	0.762	+0.284 [+0.224, +0.346]	
			2	0.721	+0.243 [+0.186, +0.299]	
Qwen3-32B	Compositional	0.277	0	0.622	+0.344 [+0.273, +0.418]	0.0046
			1	0.624	+0.347 [+0.267, +0.427]	
			2	0.588	+0.310 [+0.233, +0.388]	

Table 27: Teacher composition on the statedness-balanced pool for five models. Recall in percent, three training seeds per teacher; every other setting is held fixed. Each reduced column trains the same adapter on labels from SDA’s teacher with one signal removed. Table 28 gives paired differences with 95% intervals.

Model	Budget	SDA	Without the workspace readout	Without the output likelihood
Qwen3-8B	$k = 1$	40.5	42.1	36.7
	$k = 2$	53.7	54.8	49.8
	$k = 3$	51.8	50.6	50.5
Qwen3-14B	$k = 1$	35.5	36.2	34.3
	$k = 2$	44.1	43.4	42.4
	$k = 3$	45.3	45.8	45.2
Qwen3-32B	$k = 1$	44.9	43.4	45.7
	$k = 2$	53.3	52.2	54.7
	$k = 3$	54.8	55.3	56.6
Qwen2.5-7B	$k = 1$	17.4	16.2	17.6
	$k = 2$	33.3	32.2	32.4
	$k = 3$	39.1	37.6	37.3
OLMo-2-7B	$k = 1$	49.5	46.2	45.2
	$k = 2$	54.7	53.4	51.1
	$k = 3$	55.5	55.5	55.2

Table 28: SDA minus the student of the teacher with one signal removed on the statedness-balanced pool, for each of five models. Differences in recall are seed-averaged points with 95% intervals from 4,000 paired whole-episode bootstrap draws; three seeds per teacher and model.

Model	Teacher	$k = 1$	$k = 2$	$k = 3$
Qwen3-8B	Without workspace readout	-1.6 [-4.0, +0.5]	-1.1 [-3.7, +1.6]	+1.2 [-1.2, +3.6]
	Without output likelihood	+3.8 [+1.6, +6.2]	+4.0 [+1.9, +6.1]	+1.3 [-0.6, +3.3]
Qwen3-14B	Without workspace readout	-0.7 [-2.7, +1.2]	+0.6 [-0.6, +2.0]	-0.6 [-2.0, +0.9]
	Without output likelihood	+1.1 [-0.2, +2.5]	+1.6 [+0.1, +3.4]	+0.1 [-1.4, +1.6]
Qwen3-32B	Without workspace readout	+1.5 [+0.1, +2.9]	+1.0 [-0.1, +2.2]	-0.5 [-1.8, +1.0]
	Without output likelihood	-0.8 [-2.9, +1.1]	-1.5 [-3.1, +0.2]	-1.7 [-3.6, +0.0]
Qwen2.5-7B	Without workspace readout	+1.1 [-0.5, +2.6]	+1.1 [-0.6, +2.9]	+1.5 [+0.0, +2.9]
	Without output likelihood	-0.3 [-1.6, +1.1]	+0.9 [-1.1, +2.9]	+1.8 [+0.3, +3.5]
OLMo-2-7B	Without workspace readout	+3.3 [+1.2, +5.5]	+1.3 [-0.5, +3.2]	+0.0 [-1.9, +1.8]
	Without output likelihood	+4.3 [+2.0, +6.7]	+3.6 [+1.2, +6.1]	+0.3 [-1.9, +2.5]

Table 29: Paired-passages results for each model: 181 pairs, 362 episodes per model. Entries are SDA minus the student of the teacher with one signal removed, averaged over three seeds, in points. Recall is blind semantic answer grading; retention indicates admission of the episode’s useful concept. The 95% intervals use 4,000 bootstrap draws resampling pairs within each model.

Model	Budget	Without the workspace readout		Without the output likelihood	
		Retention	Recall	Retention	Recall
Qwen3-8B	$k = 1$	+5.3 [+2.5, +8.2]	-0.1 [-2.0, +1.8]	+2.9 [+0.0, +5.8]	+3.0 [+1.1, +4.8]
	$k = 2$	+3.1 [+0.2, +6.2]	-0.2 [-2.6, +2.1]	+2.4 [-0.2, +5.2]	+3.2 [+1.0, +5.5]
	$k = 3$	+3.5 [+1.1, +6.0]	+1.7 [-0.6, +3.8]	+2.2 [+0.1, +4.3]	+2.3 [+0.3, +4.3]
Qwen3-14B	$k = 1$	+9.0 [+6.5, +11.7]	+1.4 [-0.6, +3.5]	-1.4 [-3.5, +0.7]	+0.3 [-1.2, +1.8]
	$k = 2$	+6.0 [+3.8, +8.5]	+0.0 [-1.7, +1.8]	-1.3 [-3.3, +0.7]	+1.3 [-0.5, +3.1]
	$k = 3$	+2.3 [+0.6, +4.0]	+0.6 [-0.9, +2.0]	+1.6 [-0.1, +3.3]	+1.8 [+0.0, +3.5]
Qwen3-32B	$k = 1$	+3.8 [+2.1, +5.6]	+1.6 [+0.2, +3.0]	-10.6 [-13.3, -7.9]	-1.9 [-4.0, +0.1]
	$k = 2$	+1.8 [-0.1, +3.7]	+0.7 [-0.6, +2.0]	-6.2 [-8.0, -4.4]	-0.3 [-2.2, +1.7]
	$k = 3$	+0.7 [-0.6, +2.2]	+0.2 [-0.9, +1.2]	-4.5 [-6.3, -2.9]	-0.8 [-2.3, +0.6]
Qwen2.5-7B	$k = 1$	+8.3 [+6.3, +10.3]	+3.0 [+1.7, +4.5]	-1.6 [-4.0, +0.8]	+0.8 [-0.7, +2.5]
	$k = 2$	+6.2 [+4.5, +7.9]	+1.8 [+0.5, +3.2]	+0.6 [-1.2, +2.4]	+0.6 [-1.6, +2.6]
	$k = 3$	+4.4 [+2.8, +6.2]	+1.5 [-0.1, +3.1]	+4.3 [+2.6, +6.2]	+0.4 [-1.5, +2.1]
OLMo-2-7B	$k = 1$	+10.0 [+7.6, +12.5]	+3.4 [+1.5, +5.4]	+3.9 [+0.8, +6.8]	+4.4 [+2.4, +6.5]
	$k = 2$	+6.0 [+4.1, +7.9]	+0.7 [-1.0, +2.4]	+5.0 [+2.5, +7.5]	+3.1 [+0.6, +5.6]
	$k = 3$	+4.0 [+2.5, +5.5]	+1.7 [+0.5, +2.9]	+4.8 [+2.6, +7.1]	+3.4 [+1.3, +5.5]

Table 29 gives the paired-passages results for each model.

We evaluate 362 Qwen3-8B episodes in 181 pairs, each sharing an identical candidate set across two passages and their own questions. Each episode includes the other member’s useful concept as a candidate, so only the passage distinguishes which candidate is useful. Table 30 reports SDA recall of 38.1/49.1/52.3% at one, two and three stored concepts, with 48.2% retention at one. Removing the workspace readout from SDA’s teacher costs 5.3 retention points ([+2.4, +8.4]) and leaves recall unchanged on this set: the difference is -0.1 points ([-2.0, +1.8]). Removing the output likelihood costs both, 2.9 retention points ([+0.1, +5.7]) and 3.0 recall points ([+1.1, +4.9]). These differences are seed-averaged, with 95% bootstrap intervals over episodes from 4,000 draws. Removing the workspace readout also costs 3.1 and 3.5 retention points at two and three stored concepts, with both intervals excluding zero. The both-of-a-pair column reports 20.8% for SDA, 17.9% for its teacher without the workspace readout and 20.4% for its teacher without the output likelihood. A passage-blind candidate ordering cannot raise this rate above zero at one stored concept. Against the untrained baselines at one stored concept, SDA improves recall by 10.5 points over the yes/no importance report ([+6.7, +14.3]) and by 2.5 points over the 1–10 importance rating ([-1.0, +6.0]).

Table 30: Paired-passage evaluation on 362 Qwen3-8B episodes in 181 pairs. Each pair shares an identical candidate set across two passages with their own questions and different useful concepts; only the passage distinguishes the useful candidate. Recall is blind semantic grading of the answer, the metric used throughout the paper. Retention is the rate at which the admitted set contains the episode’s own useful concept. The last column is the rate at which both members of a pair retain their own useful concept. Rows named by a removed signal are adapters trained on SDA’s teacher with that signal removed; trained policies average three seeds; the untrained rows score candidates directly.

Admission policy	Recall			Retention	
	$k = 1$	$k = 2$	$k = 3$	$k = 1$	Both of a pair
SDA	38.1	49.1	52.3	48.2	20.8
Without the workspace readout	38.2	49.3	50.6	42.8	17.9
Without the output likelihood	35.2	45.9	50.0	45.3	20.4
1–10 importance rating (Park et al., 2023)	35.6	43.6	46.1	39.5	13.8
Yes/no importance report	27.6	38.7	43.4	18.8	3.9
Output likelihood used directly	26.8	39.5	40.6	21.6	3.3
Workspace readout used directly	22.6	34.0	37.3	17.1	5.0

Over the 220 teacher-labeled episodes, each containing a useful candidate, SDA’s teacher’s two positive labels differ from those of the teacher without the workspace readout on 43.2% of episodes and from those of the teacher without the output likelihood on 45.0%. On disagreement episodes, SDA’s teacher’s labels contain the useful concept in 63.2% against 58.9% for the teacher without the workspace readout (exact McNemar $p = 0.67$), and in 63.6% against 48.5% for the teacher without the output likelihood ($p = 0.032$). Across all 220 episodes, these rates are 60.9%, 59.1% and 54.1%, respectively.

The teacher uses each candidate’s first token as specified in Eq. (3). The full-candidate readout protocol appends the candidate and reads each token at the position preceding it. Each token receives its peak reciprocal rank over layers; the candidate score aggregates these peaks by their mean or minimum. Each aggregation and the first-token readout take their own maximum over surface forms.

On the standard Fresh-2 pool with Qwen3-8B, 56% of candidates span multiple tokens. Averaging over all candidate tokens improves ranking over the matched first-token readout (Table 31).

Table 31: Candidate readout comparison on the standard pool, Qwen3-8B, 363 episodes and 1,607 candidates. Each readout takes its own maximum over surface forms. The first-token readout is recomputed in the same measurement call as the two full-candidate aggregations.

Workspace readout	Pooled AUC	Within-episode AUC
First token	0.530	0.536
Mean across all tokens	0.574	0.571
Minimum across all tokens	0.531	0.538

Teacher composition on the standard pool. On the standard Fresh-2 pool (Table 32), the teacher without the workspace readout reaches 50.3%, a paired +0.3 percentage points relative to SDA (95% CI $[-2.2, +2.8]$); the teacher without the output likelihood reaches 47.9%, a paired -2.1 points ($[-4.1, -0.3]$).

Distillation changes the usable ranking. Using the combined teacher directly yields 45.7% standard-pool Fresh-2 recall, -4.3 points relative to SDA ($[-8.9, +0.1]$). Direct workspace scoring yields 44.6%, a difference of -5.4 points ($[-10.4, -0.5]$), and direct likelihood yields 48.8%, a difference of -1.3 points ($[-5.6, +2.8]$). The student learns a context-and-candidate judgment from the teacher’s ordering.

Table 32: SDA ablations on the standard Fresh-2 pool (Qwen3-8B, $k = 2$, blind semantic recall in percent). Trained variants average three seeds. Δ : recall difference from SDA with a paired, seed-averaged 4,000-draw episode bootstrap 95% interval. The untrained report and the workspace readout and output likelihood used directly are in Table 15.

Variant	Recall	Δ vs. SDA
SDA	50.0	<i>reference</i>
<i>Teacher with one signal removed</i>		
Without the output likelihood	47.9	-2.1 [-4.1, -0.3]
Without the workspace readout	50.3	+0.3 [-2.2, +2.8]
<i>No distillation</i>		
Combined teacher used directly	45.7	-4.3 [-8.9, +0.1]
<i>Recall reward</i>		
Recall reward instead of distillation	46.4	-3.7 [-6.2, -1.3]
SDA followed by recall-reward refinement	52.9	+2.8 [+0.6, +5.1]

Table 33: Objective ablation on the standard pool, Qwen3-8B, $k = 2$, blind semantic recall in percent. The main objective rows average three seeds at learning rate 10^{-5} ; each smaller-rate row uses one seed. Validation selects the checkpoint in every run.

Training objective	Recall
SDA: token-level cross-entropy	50.0
Yes/no log-odds cross-entropy	40.5
Yes/no log-odds, learning rate 2×10^{-6}	40.2
Yes/no log-odds, learning rate 5×10^{-6}	40.5

H.2 THE TRAINING OBJECTIVE DETERMINES ADMISSION ORDERING

The objective baseline aligns training with the admission score by applying binary cross-entropy to the yes/no log-odds in Eq. (5). Teacher labels, training episodes, adapter shape and the 414-step budget are fixed. Validation within-episode admission AUC starts at 0.657 for the untrained model and reaches 0.824 at the final step under SDA’s vocabulary-normalized token objective. At learning rate 10^{-5} , the log-odds objective ends at 0.250, 0.333 and 0.301 across the three seeds. Validation selects the untrained checkpoint in all three runs (Table 33).

At a step size five times smaller (2×10^{-6}), validation selects a trained checkpoint after 100 steps, with within-episode AUC 0.690 and standard-pool recall of 40.2% at $k = 2$, against 50.0% for SDA’s vocabulary-normalized token prediction. At 5×10^{-6} , selection returns the untrained checkpoint. Each smaller-step-size run uses one seed. SDA’s token objective yields 9.8 points more recall than the trained log-odds checkpoint and 9.6 points more than the untrained checkpoint selected at the other learning rates.

Reader reward adds task-specific supervision. SDA exceeds recall-reward training by 3.7 points on Fresh-2 ([+1.3, +6.2]); recall-reward refinement after SDA adds 2.8 points ([+0.6, +5.1]). These ablations use the same training episodes and adapter shape with a matched step budget (App. B.6).